Infectious Diseases I

Jacob Schwarz, Pharm.D., MBA, BCIDP, BCCCP, BCPS, FAzPA

Mayo Clinic

Phoenix, AZ

Infectious Diseases I

Jacob Schwarz, Pharm.D., MBA, BCIDP, BCCCP, BCPS, FAzPA

Mayo Clinic

Phoenix, AZ

Learning Objectives

1. Develop risk factor–based empiric antibiotic regimens for patients with suspected ventilator-associated pneumonia.
2. Develop empiric and definitive antimicrobial therapy plans for patients with central line-associated bloodstream infections and catheter-related urinary tract infections.
3. Differentiate between location of intra-abdominal infections and respective empiric antimicrobial therapy, including the role of antibiotics in patients with acute pancreatitis.
4. Develop a definitive management strategy for critically ill patients with severe *Clostridioides difficile* infection.
5. Describe the role of pharmacotherapy and recommend appropriate therapy for patients with severe postoperative wound infection or severe cutaneous reactions.
6. Identify a disease-specific and supportive care management plan for critically ill patients with severe influenza and novel severe acute respiratory syndrome coronavirus 2.

Abbreviations in This Chapter

ARDS	Acute respiratory distress syndrome
BAL	Bronchoalveolar lavage
CAUTI	Catheter-associated urinary tract infection
CDI	*Clostridioides difficile* infection
CLABSI	Central line–associated bloodstream infection
COVID-19	Coronavirus disease 2019
CVC	Central venous catheter
DRESS	Drug rash with eosinophilia and systemic symptoms
ECMO	Extracorporeal membrane oxygenation
ED	Emergency department
ICU	Intensive care unit
IDSA	Infectious Diseases Society of America
IVIG	Intravenous immunoglobulin
MDRO	Multidrug-resistant organism
MICU	Medical intensive care unit
MRSA	Methicillin-resistant *Staphylococcus aureus*
MRSE	Methicillin-resistant *Staphylococcus epidermidis*
MSSA	Methicillin-sensitive *Staphylococcus aureus*
MV	Mechanical ventilation
NHSN	National Healthcare Safety Network
SARS-CoV-2	Severe acute respiratory syndrome coronavirus 2
RT-PCR	Reverse transcription polymerase chain reaction
SICU	Surgical intensive care unit
SIRS	Systemic inflammatory response syndrome
SJS	Stevens-Johnson syndrome
TBSA	Total body surface area
TEN	Toxic epidermal necrolysis
UTI	Urinary tract infection
VAP	Ventilator-associated pneumonia

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

Questions 1 and 2 pertain to the following case.

K.P., a 38-year-old otherwise healthy woman, was involved in a motor vehicle collision, sustaining a traumatic brain injury and severe chest injuries requiring endotracheal intubation and mechanical ventilation. After being in the trauma intensive care unit (ICU) for 96 hours, a new infiltrate is noted on her chest radiograph, as well as a temperature of 101.9°F (38.8°C), a white blood cell count (WBC) of 15 x 10^3 cells/mm^3, and macroscopically purulent sputum. K.P. is hemodynamically stable. Accordingly, your team decides to obtain a bronchoscopic bronchoalveolar lavage (BAL) of the affected lung field to assess for ventilator-associated pneumonia (VAP), where your local ICU MRSA prevalence is less than 10%.

1. Which is the most likely causative pathogen of K.P.'s suspected VAP?
   A. *Acinetobacter* spp.
   B. Methicillin-resistant *Staphylococcus aureus* (MRSA)
   C. Multidrug-resistant *Pseudomonas aeruginosa*
   D. *Streptococcus pneumoniae*

2. Based on the 2016 IDSA guidelines, which empiric antibiotic regimen is best for the likely causative pathogen(s) of K.P.'s suspected VAP?
   A. Azithromycin plus moxifloxacin
   B. Cefepime
   C. Ceftriaxone
   D. Ceftriaxone plus vancomycin

3. T.D. is a 57-year-old man admitted to the medical intensive care unit (MICU) with severe hyperosmolar hyperglycemic syndrome. A subclavian central venous catheter (CVC) is placed on his arrival at the MICU. On ICU day 4, T.D. has a temperature of 102.7°F (39.2°C) and a WBC of 17 x 10^3 cells/mm^3. The nurse notes new-onset erythema at the catheter site. After an appropriate diagnostic workup, it is decided to initiate empiric antibiotic therapy for a suspected central line–associated bloodstream infection (CLABSI). Which agent is best for empiric therapy?
   A. Cefazolin
   B. Linezolid
   C. Piperacillin/tazobactam
   D. Vancomycin

4. D.G., a 31-year-old woman, presents to the MICU with severe respiratory failure and acute respiratory distress syndrome (ARDS) requiring intubation after 48 hours of malaise, fever, and myalgias. The nasal washing sent by the emergency department (ED) for rapid diagnostic testing is positive for influenza A. The local prevalence of subtype 2009 H1N1 is high. Which agent would be most appropriate for initial treatment of D.G.'s severe influenza?
   A. Amantadine
   B. Inhaled zanamivir
   C. Intravenous zanamivir
   D. Oseltamivir

5. T.S. is a 79-year-old woman admitted to the MICU for respiratory failure and severe community-acquired pneumonia. T.S. has had a urethral catheter in place for 6 days while mechanically ventilated on fentanyl infusion and intermittent haloperidol as needed for pain and delirium, respectively. This morning, T.S. had a temperature of 101.6°F (38.7°C) and an elevation in WBC to 16 x 10^3 cells/mm^3; she is hemodynamically stable. Blood and urine cultures are sent. Urinalysis reveals significant pyuria. Which pathogen is most likely to cause a catheter-associated urinary tract infection (CAUTI) in T.S.?
   A. *Enterococcus faecalis*
   B. *Escherichia coli*
   C. *P. aeruginosa*
   D. *S. aureus*

6. K.D. is a 59-year-old man admitted to the surgical intensive care unit (SICU) after an emergency operation and partial bowel resection with primary anastomosis for mid-small bowel necrosis and perforation likely secondary to severe peripheral vascular disease. During the operation, significant peritoneal contamination with evidence of gross peritonitis was noted together with persistent hypotension and need for vasopressors. K.D. received perioperative cefazolin and metronidazole. Which empiric antibiotic regimen would be most appropriate for K.D.?
   A. Ceftriaxone and vancomycin
   B. Ciprofloxacin and metronidazole
   C. Ertapenem
   D. Piperacillin/tazobactam

7. T.M. is a 42-year-old man with chronic alcoholism who presents to the ED with severe epigastric pain and serum lipase greater than 10 times the upper limit of normal. The resident orders a computed tomography (CT) scan, which reveals necrosis affecting almost 40% of the pancreas but no abnormal fluid collections or evidence of abscess. T.M. is febrile and tachycardic, and his WBC is elevated. T.M.'s urine output is less than 0.5 mL/kg/hour, suggestive of hypovolemia. Which best describes the role of antibiotic therapy for T.M. at this time?
   A. Antibiotic therapy is not indicated.
   B. Initiate empiric antibiotic therapy for presumed sepsis and likely pancreatic infection.
   C. Initiate perioperative antibiotic therapy in preparation for pancreatic debridement.
   D. Initiate prophylactic antibiotic therapy to prevent infection of necrotic tissue.

8. J.E. is a 67-year-old woman admitted to the MICU for severe metabolic acidosis secondary to unintentional metformin overdosage. Her ICU stay has been complicated by a femoral vein CLABSI and related severe sepsis caused by pan-sensitive *E. coli*. J.E. received a 3-day course of empiric piperacillin/tazobactam, removal of her central line, and 11 days of ceftriaxone as definitive therapy with resolution of sepsis. Starting yesterday, it was noted that J.E. had nine loose bowel movements and new leukocytosis of 14,500 cells/mm^3, which suggests *Clostridioides difficile* infection (CDI). J.E. continues to tolerate enteral nutrition. Per the 2021 IDSA focused guideline update, which is the most appropriate regimen for J.E.'s suspected CDI?
   A. Fidaxomicin 200 mg per feeding tube every 12 hours
   B. Metronidazole 500 mg per feeding tube every 8 hours
   C. Metronidazole 500 mg intravenously every 8 hours
   D. Vancomycin 125 mg per feeding tube every 6 hours

9. J.S. is a 42-year-old man admitted to the SICU after an open total colectomy with ileostomy for ischemic colitis. On postoperative day 4, the surgery resident on your interdisciplinary ICU team notes moderate erythema (3 cm) and purulent drainage from the wound. The resident subsequently opens the skin portion of the wound and finds infected material just above the unaffected fascia. No systemic signs and symptoms are noted. Which intervention is most appropriate?
   A. Continue dressing changes and patient assessment.
   B. Initiate empiric vancomycin for postoperative wound infection.
   C. Initiate empiric antibiotic therapy targeted against skin and colonic flora.
   D. Initiate broad-spectrum antibiotic therapy for suspected necrotizing fasciitis.

10. A.C. is a 27-year-old man with human immunodeficiency virus (HIV) on active antiretroviral therapy who presents to the burn ICU from an outside hospital with 30% total body surface area (TBSA) epidermolysis of his back and upper arms suggestive of toxic epidermal necrolysis (TEN), likely from sulfamethoxazole/trimethoprim. Reportedly, A.C. arrived at the outside hospital about 6 hours ago with less than 10% TBSA involvement. Vital signs include heart rate 142 beats/minute, respiratory rate 30 breaths/minute, and blood pressure 100/50 mm Hg. Which would be the best initial pharmacotherapeutic intervention?
   A. Administer high-dose corticosteroids to halt the progression of TEN.
   B. Initiate crystalloid resuscitation for hypovolemia.
   C. Place a nasogastric tube to administer antiretroviral medications.
   D. Initiate broad-spectrum antibiotic prophylaxis for wound infection.

11. T.F. is a 68-year-old man with diabetes and hypertension who presents to the ED with altered mental status and acute kidney injury. T.F. was seen at an outside hospital 5 days ago, where he tested positive for novel severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). On initial assessment, T.F. is hemodynamically stable with an oxygen saturation (SaO_2) of 87% on room air, which has worsened over the past 4 hours. T.F. now requires 5 L of high-flow nasal cannula to maintain an SaO_2 above 92%. Chest radiography is notable for bilateral opacities consistent with viral pneumonia. Initial laboratory tests indicate alanine aminotransferase 23 U/L and estimated creatinine clearance 31 mL/minute (baseline 68 mL/minute). Which agent is considered first-line therapy in hospitalized patients with coronavirus disease 2019 (COVID-19) and increasing oxygen requirement?
   A. Dexamethasone.
   B. Hydroxychloroquine.
   C. Remdesivir.
   D. Tocilizumab.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task B: 5
2. Domain 2: Therapeutics and Patient Managment
   a. Task A: 1
   b. Task B: 7, 12
3. Domain 3: Professional Practice
   a. Task B: 1

I. VENTILATOR-ASSOCIATED PNEUMONIA

A. Epidemiology
1. About 90% of hospital-acquired pneumonia episodes in critically ill patients occur during mechanical ventilation. Mechanical ventilation is an independent risk factor for pneumonia.
2. Between 9% and 27% of mechanically ventilated patients develop pneumonia.
3. The incidence of pneumonia increases with the duration of mechanical ventilation, with the highest risk during the first 5 days. Individual risk factors for VAP include underlying chronic lung disease, acute lung injury, aspiration, coma, trauma (e.g., chest trauma; traumatic brain injury), burns, reintubation, and overall severity of illness.
4. VAP accounts for more than 50% of antibiotic use in critically ill patients and has an attributable cost of more than $40,000 per episode.
5. VAP has an estimated attributable mortality of 10%–50%, which varies among critically ill populations.

B. Classification and Etiology
1. VAP is a distinct type of hospital-acquired pneumonia that occurs more than 48 hours after endotracheal intubation. VAP is further classified by the Centers for Disease Control and Prevention (CDC) as a ventilator-associated event and infection-related ventilator-associated complication. Updated guidelines from the IDSA for the diagnosis and management of VAP were published in 2016. (Please see Table 2 at the end of this section for summary and comparison with 2005 guidelines.)
2. VAP is usually caused by bacterial pathogens; may be monomicrobial or polymicrobial, but is rarely caused by viral or fungal pathogens.
3. The prevalence of specific bacterial pathogens varies between critically ill patient populations, geographic region, and local antibiotic usage patterns. Risk factors for multidrug-resistant organisms (MDROs) are summarized in Box 1.
4. Classification of VAP as early onset (within the first 2–4 days of hospitalization) or late onset (after 5 days or more of hospitalization) to differentiate between suspected pathogens is no longer recommended. However, longer duration (e.g., 5 days) of hospitalization before the onset of VAP is associated with nosocomial and MDRO pathogens.

Box 1. Risk Factors for MDROs Causing VAP

Prior intravenous antibiotic therapy in preceding 90 days
Acute renal replacement therapy before VAP onset
5 or more days of hospitalization before the occurrence of VAP
Septic shock at the time of VAP
ARDS preceding VAP

MDRO = multidrug-resistant organism.

C. Prevention
1. VAP is considered a reportable and preventable complication of endotracheal intubation and mechanical ventilation.
2. Data are emerging on effective prevention strategies.
3. Recommended essential practices for preventing VAP include (Infect Control Hosp Epidemiol 2022; 43:687-713):
   a. Avoid intubation and prevent reintubation if possible.
   b. Minimize sedation of ventilated patients whenever possible.
      i. Preferentially use multimodal strategies and medications other than benzodiazepines to manage agitation.

ii. Use a protocol to minimize sedation.
iii. Implement a ventilator liberation protocol.
c. Maintain and improve physical conditioning, provide early exercise and mobilization.
d. Elevate head of bed to 30–45 degrees.
e. Maintain integrity of mechanical ventilator circuit.
f. Provide oral care with toothbrushing but without chlorhexidine.
g. Avoid oral care with chlorhexidine to prevent VAP in patients per recent joint recommendations by SHEA/IDSA/APIC. Meta-analyses have shown no benefit and the potential of higher mortality rates for patients receiving chlorhexidine oral care.
h. May consider use of selective oropharynx and digestive decontamination in ICUs with low prevalence of antibiotic resistance organisms.
i. Avoid the use of probiotics to prevent VAP in mechanically ventilated patients. While some meta-analyses found there might be a benefit, a recent large RCT found no difference in the development of VAP in mechanically ventilated patients (Crit Care Med 2022;50:1175-86; JAMA 2021;326:1024-33).

D. Diagnosis
1. The diagnostic approach for VAP includes (1) determining whether clinical signs and symptoms are caused by pneumonia and (2) if pneumonia is present, identifying the causative pathogen(s), preferably using lower respiratory tract culture.
2. About half of mechanically ventilated patients with a clinical suggestion of pneumonia will have bacteriologically confirmed pneumonia.
3. Clinical signs and symptoms of pneumonia include new or changing infiltrate on chest radiograph and at least two of the following:
   a. Elevated WBC
   b. Fever (e.g., temperature greater than 100.4°F [38°C])
   c. Macroscopically purulent sputum production
   d. Impaired or worsening oxygenation
4. Lower respiratory tract cultures can be obtained through noninvasive or invasive techniques and reported as qualitative, semiquantitative, or quantitative.
   a. Quantitative (expressed as CFU/mL) and semiquantitative (expressed as rare/few to many) cultures using recommended diagnostic growth thresholds are more specific than qualitative cultures for identifying the causative pathogen.
   b. Noninvasive techniques: Tracheal aspirate from endotracheal or tracheostomy tube – proximal to distal sampling (depending on depth of sample) of upper airway secretions; usually semiquantitative
   c. Invasive techniques
      i. Blind, catheter-directed, or bronchoscopic BAL – Distal sampling of lung lobe/segment using saline lavage; significant quantitative growth threshold above 10,000 or 100,000 CFU/mL
      ii. Bronchoscopic protected specimen brush (PSB) – Distal sampling of specific bronchial segment; significant quantitative growth threshold above 1000 CFU/mL
   d. Rapid diagnostic tools can also be used to help in the identification of potential pathogens more speedily. Please see the Infectious Diseases II chapter for further discussion on the use of rapid diagnostic tests in the ICU.
5. According to the current IDSA guidelines, the suggested diagnostic strategy for VAP includes clinical suspicion and use of noninvasive sampling with semiquantitative cultures to diagnose VAP, rather than invasive sampling or noninvasive sampling with quantitative cultures.
   a. If an invasive strategy with quantitative cultures is used, it is suggested that antibiotics be withheld rather than continued if quantitative culture results are below the diagnostic threshold for VAP (i.e., PSB less than 1000 CFU/mL; BAL less than 10,000 CFU/mL).

E. Treatment
   1. Antibiotic therapy for VAP is empiric or definitive.
   2. Antibiotic selections should include intravenous agents able to achieve relevant pulmonary tissue concentrations related to pathogen minimum inhibitory concentration (MIC) dosed optimally using evidence-based pharmacokinetic and pharmacodynamic principles.
   3. Empiric antibiotic therapy should be initiated in patients with clinical suspicion for VAP (Table 1). Patients with septic shock should receive antibiotics within 1 hour from onset of hypotension (see Sepsis chapter).
      a. Inappropriate empiric antibiotic therapy for VAP (i.e., delay in or absence of antibiotics active against identified causative pathogen[s]) is associated with increased mortality.
      b. To decrease the likelihood of inappropriate therapy, empiric antibiotic selections should be based on:
         i. Presence of MDRO risk factor
         ii. Local VAP pathogen prevalence, particularly MRSA
         iii. Local antibiotic susceptibility patterns (i.e., ICU-specific antibiogram)
      c. All patients with suspected VAP should receive an antibiotic active against MSSA, *P. aeruginosa*, and other gram-negative bacilli.
         i. Suggest prescribing two antipseudomonal antibiotics from different classes in patients with any of the following:
            (a) MDRO risk factor
            (b) ICU antibiogram with greater than 10% of gram-negative isolates resistant to an agent being considered for monotherapy (i.e., multidrug-resistant [MDR] *P. aeruginosa*)
            (c) ICU where local antimicrobial susceptibility rates unavailable
      d. Suggest including an agent active against MRSA in patients with any of the following:
         i. MDRO risk factor
         ii. ICU MRSA prevalence greater than 10%–20% of *S. aureus*. (Note: MRSA rates vary between centers, ICUs, and populations, with general rates in U.S. hospitals greater than 20% [https://resistancemap.onehealthtrust.org/AntibioticResistance.php].)
         iii. Prevalence of MRSA is not known
      e. Combination antibiotic therapy using agents with similar bacterial spectra but different mechanisms of action may be necessary to increase the likelihood of appropriate empiric antibiotic therapy for other gram-negative pathogens.
      f. Lower respiratory tract cultures should be obtained before initiation of antibiotic therapy to increase the likelihood of identifying causative pathogen(s). Inappropriate delays in initiation of antibiotic therapy should be avoided in unstable patients (e.g., patients with septic shock).
      g. Empiric antibiotic therapy should be de-escalated to definitive therapy, depending on identified pathogen(s) and antibiotic susceptibilities. The GRACE-VAP trial, a recently published non-inferiority trial evaluated the use of Gram stain-guided restrictive antibiotic therapy in treatment of VAP. The authors found that Gram stain-guided was non-inferior to guideline-directed therapy in clinical response (76.7% vs. 71.8%), 28-day mortality (13.6% vs. 17.5%), and similar ICU-free days, ventilator-free days, and adverse events (JAMA Netw Open 2022;5:e226136).

Table 1. VAP Classifications and Recommended Empiric Antibiotic Therapy Based on 2016 IDSA VAP Guidelines

Likely Pathogens	Recommended Empiric Antibiotic(s)[a]
Patients without MDRO risk factor: Single antipseudomonal agent with 90% or greater empiric activity on local ICU antibiogram, and local MRSA prevalence is less than 10-20%	
P. aeruginosa *S. pneumoniae* *Haemophilus influenzae* α- and β-hemolytic *Streptococcus* spp. Methicillin-sensitive *S. aureus* (MSSA) Antibiotic-sensitive enteric gram-negative bacilli (GNB) (e.g., *E. coli*; *Klebsiella* spp.; *Enterobacter* spp.; *Proteus* spp.; *Serratia* spp.)	Cefepime, imipenem, levofloxacin, meropenem, piperacillin/tazobactam
Patient without MDRO risk factor: Single antipseudomonal agent with less than 90% empiric activity on local ICU antibiogram and/or local MRSA prevalence greater than 10-20%[b]	
P. aeruginosa Methicillin-resistant *S. aureus* (MRSA) *S. pneumoniae* *Haemophilus influenzae* α- and β-hemolytic *Streptococcus* spp. Methicillin-sensitive *S. aureus* (MSSA) Antibiotic-sensitive enteric gram-negative bacilli (GNB) (e.g., *E. coli*; *Klebsiella* spp.; *Enterobacter* spp.; *Proteus* spp.; *Serratia* spp.)	Double Antipseudomonal Coverage Antipseudomonal β-lactam[c] (aztreonam, cefepime, ceftazidime, imipenem, meropenem, piperacillin/tazobactam) + Antipseudomonal fluoroquinolone (ciprofloxacin or levofloxacin) OR aminoglycoside[d] (amikacin or tobramycin) MRSA Linezolid or vancomycin
Patients with MDRO risk factor	
P. aeruginosa Methicillin-resistant *S. aureus* (MRSA) *Acinetobacter* spp. Antibiotic-resistant enteric GNB (e.g., extended-spectrum β-lactamase [ESBL]-producing organisms) *Stenotrophomonas maltophilia*	Antipseudomonal β-lactam[c] (aztreonam, cefepime, ceftazidime, imipenem, meropenem, piperacillin/tazobactam) + Antipseudomonal fluoroquinolone (ciprofloxacin or levofloxacin) OR aminoglycoside[d] (amikacin, gentamicin, or tobramycin) AND Linezolid or vancomycin

[a]Agents/classes are listed in alphabetic order.

[b]There may be situations in which a patient has no MDRO risk factors but may meet the recommendation to receive dual antipseudomonal therapy and/or MRSA coverage.

[c]Doripenem is no longer available in the United States. Before being discontinued, doripenem had a U.S. Food and Drug Administration (FDA) warning against use for treatment of ventilator-associated bacterial pneumonia.

[d]Guidelines suggest avoiding empiric aminoglycoside if alternative agents with activity are available. New CLSI guidance have lowered MIC breakpoints and no longer recommend the use of gentamicin in the treatment of *P. aeruginosa* infections.

4. Definitive antibiotic therapy should be focused on the causative pathogen(s) identified on lower respiratory tract culture.
    a. Pathogen-specific definitive antibiotic choices should be based on antibiotic susceptibilities and, if possible, available evidence supporting efficacy and safety in patients with VAP.
        i. MRSA: Vancomycin or linezolid; daptomycin is not indicated for pneumonia because of direct inhibition by pulmonary surfactant. Although not included in the guidelines, ceftaroline is also a potential treatment of MRSA HAP/VAP.
        ii. *P. aeruginosa*
            (a) Monotherapy with a β-lactam listed below and found to be susceptible on final culture results is recommended in immunocompetent patients not in septic shock and not at a high risk of dying. Concomitant aminoglycoside or antipseudomonal fluoroquinolone is recommended in immunocompromised patients and those in septic shock or at a high risk of dying. Combination therapy can be de-escalated to monotherapy β-lactam once septic shock resolves.
            (b) β-Lactams: Cefepime, ceftazidime (has minimal activity against MSSA and should not be used to for both *P. aeruginosa* and MSSA), piperacillin/tazobactam, imipenem, or meropenem
            (c) Aminoglycosides: Amikacin, gentamicin, or tobramycin
            (d) Fluoroquinolones: Ciprofloxacin or levofloxacin
        iii. *Acinetobacter* spp.: Ampicillin/sulbactam (sulbactam is active agent) or carbapenem (e.g., imipenem or meropenem)
        iv. *Stenotrophomonas maltophilia*: Sulfamethoxazole/trimethoprim, levofloxacin, minocycline, or combination of double-coverage in critical/serious infections
        v. Extended-spectrum β-lactamase (ESBL)-producing organisms: Carbapenem (e.g., ertapenem; imipenem; meropenem)
        vi. Since the 2016 IDSA guidelines, cefiderocol, ceftolozane/tazobactam, ceftazidime/avibactam, imipenem/cilastatin/relebactam, and sulbactam/durlobactam have been FDA approved for the treatment of VAP. These agents can be considered for certain multidrug-resistant (MDR) pathogens (e.g., ESBL; specific carbapenemases) in concert with an antimicrobial stewardship program.
        vii. In 2023, the IDSA updated its guidance document in the treatment of antimicrobial resistant gram-negative infections (Clin Infect Dis 2023:ciad428).
    b. Duration of definitive antibiotic therapy is recommended to be 7 days for all patients rather than longer durations. Pathogen-based recommendations for duration of definitive antibiotic are no longer indicated in VAP guidelines.
        i. The PneumA trial was a pivotal noninferiority study that evaluated 8 days (7 full treatment days) versus 15 days (14 full treatment days) for duration of definitive antibiotic therapy in patients with bacteriologically diagnosed VAP. The primary noninferiority outcome was VAP recurrence. All patients included in the study received appropriate empiric antibiotic therapy (JAMA 2003;290:2588-98). Major study findings were:
            (a) Definitive antibiotic therapy for 8 days was noninferior to 15 days in patients with VAP not caused by non–lactose-fermenting gram-negative bacilli (e.g., *P. aeruginosa*).
            (b) Patients with VAP caused by non–lactose-fermenting gram-negative bacilli in the 8-day group more often had VAP recurrence. This was primarily seen in patients with *P. aeruginosa*, but also included other lactose-non-fermenters (e.g., *Acinetobacter, Stenotrophomonas*, and *Burkholderia* spp.).
            (c) The rate of MDR pathogens during a recurrent VAP episode was higher in patients in the 15-day group.

ii. Two meta-analyses that included the results of the PneumA trial suggest limited benefit of prolonged duration of definitive antibiotic therapy for treating VAP (Chest 2013;144:1759-67; Cochrane Database Syst Rev 2015;8:CD007577). Synopsis includes:
   (a) Any increase in VAP recurrence rate is small;
   (b) Mortality and clinical cure do not appear to be affected by shorter durations;
   (c) Evidence for recurrence from subgroup analyses has important limitations.

iii. Guidelines acknowledge that situations may exist in which shorter or longer durations of definitive antibiotic therapy may be indicated, depending on the rate of clinical and radiologic improvement. A single-center, retrospective study suggested that patients with "possible VAP" who had minimal and stable ventilator settings at diagnosis are appropriate for short-course therapy (i.e., 1-3 days) (Clin Infect Dis 2017;64:870-6). However, application of these findings is limited, given that only 39% of short-course patients had an identified pathogen on tracheal aspirate or BAL culture, bringing into question the likelihood of confirmed pneumonia.

iv. The iDIAPASON Trial was a randomized, controlled, open-label trial that compared 8 versus 15 days of antibiotic therapy solely for the treatment of Pseudomonas aeruginosa ventillator-associated pneumonia (PA-VAP). The primary noninferiority endpoint was a composite outcome of death and PA-VAP recurrence during the ICU stay until day 90. Duration of therapy was counted from the time of effective antibiotic therapy (Intensive Care Med 2022;48:841-9). Major study findings were the following:
   (a) Short duration was not associated with an increased mortality, longer duration of mechanical ventilation, or length of ICU stay.
   (b) While not significant, there was a trend toward higher proportion of recurrence in the 8 days of therapy compared with 15 days (17% vs. 9.2%; confidence interval [CI], –0.5% –16.8%).

5. Response to antibiotic therapy should be assessed serially using clinical signs and symptoms of infection (e.g., oxygenation requirements, blood pressure, WBC, temperature) and status of VAP-related organ dysfunction.
   a. Lack of response in signs and symptoms of VAP beyond treatment day 3 necessitates reassessment of diagnosis, causative pathogen(s), antibiotic regimen, and presence of pneumonia-related complications (e.g., ARDS).
   b. Patients thought to have persistent VAP should undergo repeat lower respiratory tract culture and receive empiric antibiotic therapy considering previous pathogen(s) and antibiotic exposure.

Table 2. Summary of Key Changes in 2016 IDSA VAP Guidelines

Characteristic	2005 Guidelines	2016 Guidelines
Methodology	Expert opinion based on level of evidence ranging from Level I (high) to III (low)	GRADE criteria to identify "recommended" (strong) or "suggested" (weak) guidance based on level of evidence categories "very low," "low," "moderate," and "high quality"
Diagnosis	Clinical strategy or bacteriologic strategy	Suggest clinical suspicion with noninvasive sampling and semiquantitative cultures If invasive sampling is used, suggest that antibiotics be withheld rather than continued if quantitative culture results are below the diagnostic threshold for VAP

Table 2. Summary of Key Changes in 2016 IDS VAP Guidelines *(continued)*

Characteristic	2005 Guidelines	2016 Guidelines
Classification	Early onset or late onset	No longer differentiate VAP episodes based on time from hospital admission Emphasis on local risk assessment for *P. aeruginosa* and MRSA occurring early in hospitalization
MDRO Risk Factors	• Antibiotic therapy in preceding 90 days • Chronic dialysis within 30 days • Current hospitalization of 5 days or more • Home chronic wound care • Home infusion therapy • Hospitalization in the preceding 90 days • Immunosuppressive disease and/or therapy • Known contact or colonization with MDROs • Residence in a nursing home or extended-care facility	• Prior intravenous antibiotic therapy in preceding 90 days • Acute renal replacement therapy before VAP onset • 5 or more days of hospitalization before the occurrence of VAP • Septic shock at the time of VAP • ARDS preceding VAP
Empiric Therapy	Categorized based on early onset or late onset and presence of MDRO risk factor • Early onset without MDRO risk factor: no antipseudomonal or MRSA coverage • Late onset or early onset with MDRO risk factor: dual antipseudomonal and MRSA coverage	Recommend coverage for MSSA, *P. aeruginosa*, and other gram-negative bacilli in all empiric regimens. Suggest including an agent active against MRSA if any of the following: • MDRO risk factor for antimicrobial resistance • ICU MRSA prevalence greater than 10%-20% of all S. aureus • Prevalence of MRSA is not known Suggest prescribing dual antipseudomonal antibiotics from different classes if any of the following: • MDRO risk factor for antimicrobial resistance • ICUs with greater than 10% of gram-negative isolates are resistant to an agent being considered for monotherapy • ICU where local antimicrobial susceptibility rates unavailable
Aminoglycosides	No specific recommendation	Suggest avoiding if alternative agents with adequate gram-negative activity are available
Colistin	No specific recommendation	Suggest avoiding if alternative agents with adequate gram-negative activity are available
Duration of Definitive Therapy	In patients who received appropriate empiric antibiotic therapy: • Non–lactose-fermenting Gram negative bacilli: 14 days • All other pathogens: 7 days	Recommend 7 days rather than longer durations regardless of pathogen Suggest using procalcitonin plus clinical criteria to guide discontinuation of antibiotic therapy, rather than clinical criteria alone

Patient Cases

1. C.T. is a 48-year-old man sustaining a 40% TBSA burn to his left side with likely inhalational injury requiring mechanical ventilation. C.T. has been in the ICU for 7 days. Overnight, C.T. had a temperature of 102.1°F (39°C), a WBC of 18 x 10^3 cells/mm^3, and an increase in macroscopically purulent sputum production; C.T. is hemodynamically stable with no signs of new-onset organ dysfunction. C.T.'s chest radiograph is difficult to assess, given his inhalational injury. The respiratory therapist was asked by the fellow to do a catheter-directed BAL for suspected VAP. The culture was sent to the microbiology laboratory. Which would be best to do at this point?
   A. Initiate broad-spectrum empiric antibiotic therapy for suspected MDR VAP.
   B. Await 1–2 days for Gram stain results to determine empiric antibiotic regimen.
   C. Await preliminary quantitative culture results before initiating empiric antibiotic therapy.
   D. Send blood and urine cultures, and await their results before initiating empiric antibiotic therapy.

2. C.P. is a 67-year-old woman with a medical history significant for chronic obstructive pulmonary disease (COPD), type 2 diabetes, and coronary artery disease. She is readmitted to the ICU with respiratory failure requiring intubation secondary to severe COPD. C.P. had a 10-day hospitalization 12 days ago, during which time she received intravenous azithromycin for a COPD exacerbation. On day 4 of this readmission, a worsening infiltrate is in the left lower lung base with increased sputum production from admission, maximum temperature is 101.9°F (38.8°C), and there is worsening oxygenation, despite previous improvement in initial COPD exacerbation. The patient is thought to have clinical VAP, and a semiquantitative tracheal aspirate is sent to identify causative pathogen(s). The local ICU prevalence of MRSA is 30%, and the most active β-lactam antibiotic against *P. aeruginosa* on the antibiogram has 80% activity. Which empiric antibiotic regimen is best for this patient?
   A. Azithromycin plus moxifloxacin.
   B. Cefepime plus tobramycin, and vancomycin.
   C. Ceftriaxone plus azithromycin.
   D. Linezolid plus tobramycin.

3. K.L., a 37-year-old man who presents after a motorcycle collision, has sustained several orthopedic and chest injuries. K.L. has been mechanically ventilated for 8 days, during which time he was given a diagnosis of VAP caused by *Klebsiella pneumoniae*. K.L. received appropriate empiric antibiotic therapy and has improved oxygenation, decreased WBC temperature curve. Which would be the best duration of definitive antibiotic therapy for K.L.'s VAP?
   A. 24 hours after resolution of clinical signs and symptoms.
   B. 7 days.
   C. 10 days.
   D. 14 days.

II. CENTRAL LINE–ASSOCIATED BLOODSTREAM INFECTIONS

A. Epidemiology
1. Critically ill patients commonly require short-term (less than 14 days per catheter) or temporary (non-tunneled or non-implanted) placement of a CVC or central line for medication administration and hemodynamic support and monitoring.
2. More than 30,000 bloodstream infections a year are associated with the placement and use of central catheters. The number of cases has decreased over the past decade because of increased awareness and many systematic prevention efforts.
3. In 2012, the NHSN reported rates of CLABSI across adult critically ill populations of 0.8–3.4 episodes per 1000 central line–days. Aggregated national data from 2016 showed an almost 50% decrease in CLABSI rates from 2008 (0.56 vs. 1.00 cases per 1000 central line-days). Although specific rates vary between sites and populations, all central line types and locations of insertion have an attributable risk of bloodstream infection.
4. Established risk factors for CLABSI include:
   a. Excessive manipulation of the catheter during and after insertion
   b. Internal jugular or femoral insertion site
   c. Microbial colonization at the insertion site or catheter hub
   d. Neutropenia
   e. Prolonged duration of catheterization
   f. Prolonged hospitalization before catheterization
   g. Total parenteral nutrition
5. The attributable cost of a CLABSI is up to $40,000 per episode.

B. Definitions
1. Central or peripheral venous catheters are defined by the location at which the distal tip of the catheter terminates. CVCs terminating in a great vessel are considered part of the central blood circulation. The distal tip of a CVC usually resides in the inferior, superior, or distal vena cava, right atrium, or pulmonary artery (e.g., pulmonary artery catheter).
2. CVCs are usually inserted into the central venous system using the internal or external jugular, subclavian, femoral, or iliac veins. There are several types of short-term CVCs:
   a. Single- and multiple-lumen (e.g., triple lumen) catheters: Most commonly used CVC
   b. Catheter introducer: Used for massive resuscitation or facilitation of pulmonary artery catheter insertion
   c. Peripherally inserted central catheter (PICC): Short- or medium-term central catheter inserted in a peripheral vein (e.g., cephalic, basilic, or brachial veins)
   d. Pulmonary artery catheter: A central catheter of around 100 cm used for invasive hemodynamic monitoring
3. Primarily for surveillance, the CDC defines CLABSI as a primary laboratory-confirmed bloodstream infection occurring no sooner than 2 calendar days from catheter placement and no later than the day after catheter removal. Laboratory-confirmed bloodstream infection is defined as either:
   a. A recognized pathogen (i.e., not a common commensal organism) cultured from one or more blood cultures, and the organism cultured from blood is not related to an infection at another site, or
   b. A common commensal organism (e.g., diphtheroids, *Bacillus* spp., coagulase-negative staphylococci, viridans streptococci) cultured from two or more blood cultures drawn on separate occasions, and the organism cultured from blood is not related to an infection at another site, and patient has at least one of the following signs or symptoms: fever (temperature greater than 100.4°F [38°C]), chills, or hypotension

4. IDSA defines a catheter-related bloodstream infection minimally as bacteremia or fungemia in a patient who has an intravascular device and:
    a. More than one positive blood culture obtained from a peripheral vein. A definitive diagnosis requires that the same organism(s) grow from at least one percutaneous blood culture and from the CVC tip or that two blood cultures, one from a catheter hub and one from a peripheral vein, are positive for the same organism(s).
    b. Clinical manifestations of infection (e.g., fever, chills, and/or hypotension), and
    c. No apparent source for bloodstream infection other than the catheter

C. Etiology
1. CLABSIs are usually monomicrobial. Pathogen prevalence is based on patient-specific risk factors and underlying illness.
2. Common organisms responsible for CLABSIs include coagulase-negative staphylococci (e.g., *Staphylococcus epidermidis*), *S. aureus*, *Candida* spp., and enteric gram-negative bacilli (e.g., *E. coli*, *Klebsiella* spp., *Enterobacter* spp.).
3. Risk factors for MDROs include critical illness, femoral catheter placement, immunosuppression, and previous antibiotic exposure.
4. *Candida* spp. are more common in patients with the following risk factors: total parenteral nutrition, prolonged exposure to broad-spectrum antibiotics, hematologic malignancy, stem cell or solid-organ transplantation, femoral site of catheterization, or colonization owing to *Candida* spp. at several sites.

D. Prevention
1. CLABSI is considered a preventable complication. The NHSN, using CDC definitions, provides population-specific event rates for institution surveillance and performance benchmarks.
2. The foundation for preventing CLABSI includes training and education, proper aseptic insertion techniques, and active surveillance and performance improvement systems.
3. Updated recommendations for best practices in the prevention of CLABSI have been sponsored by the Society for Healthcare Epidemiology (SHEA), having been a collaborative effort between SHEA, IDSA, the Association for Professional in Infection Control (APIC), the American Hospital Associations (AHA), and the Joint Commission. These updates include evidence-based recommendations categorized as *essential practices* that should be adopted by all acute-care hospitals, or *additional approaches* that can be considered for use in locations and/or populations within hospitals when CLABSIs are not controlled after implementation of essential practices (Infect Control Hosp Epidemiol 2022;43:553-69).
    a. Essential practices
        i. Minimize central venous line insertions and durations of insertion.
        ii. Provide comprehensive education to and ensure competency for all involved with insertion, care, and maintenance of CVC.
        iii. Daily chlorhexidine baths for patients to reduce colonization
        iv. Use of a systematic process or checklist at the time of insertion to ensure adherence to proper insertion technique
        v. Alcohol-based chlorhexidine skin cleanser for skin preparation
        vi. Perform hand hygiene during insertion, care, and maintenance of the catheter
        vii. Cleanse catheter hubs before accessing.
        viii. Routine post-insertion site care
        ix. Routine replacement of administration sets not used for blood, blood products, or lipid formulations can be performed at intervals up to 7 days.
        x. Chlorhexidine-containing site/catheter dressings

b. Additional approaches
   i. Use antiseptic or antimicrobial-impregnated catheters.
   ii. Use antimicrobial ointments for hemodialysis catheter insertion sites
   iii. Use antiseptic hub cover or port protector.
   iv. Antimicrobial lock therapy
   v. Use recombinant tissue-plasminogen activating factor (rt-PA) weekly if dialysis is performed through CVC.

E. Diagnosis
1. Clinical signs and symptoms of infection have poor sensitivity and specificity. Fever is the most sensitive clinical finding, whereas inflammation or purulence at the insertion site is the most specific.
2. If a CLABSI is suspected, paired blood cultures drawn from the catheter (at least one hub/port) and from a peripheral vein should be obtained. The individual bottles should be appropriately marked through the culture-reporting period.
   a. If a blood culture cannot be drawn from a peripheral vein, it is recommended that at least two blood cultures be obtained through different catheter hubs/ports.
   b. Blood cultures positive for *S. aureus*, coagulase-negative staphylococci, or *Candida* spp. that are not attributable to another source should increase the suggestion of CLABSI.
   c. Blood cultures should be obtained before initiation of antimicrobial therapy, as appropriate.
3. A definitive diagnosis of CLABSI requires positive percutaneous blood culture results with positive culture of same pathogen from the catheter tip or catheter-drawn cultures. Please see the Infections Diseases II chapter for further discussion on the pathway for regulatory reporting of CLABSI.

F. Treatment
1. Catheter removal should be considered in all patients with a confirmed CLABSI. If a CVC is still necessary, a different anatomic site should be used. Changing to a new catheter at the same site using a guidewire or catheter introducer should be avoided.
2. Antimicrobial therapy for CLABSI is empiric or definitive.
3. Inappropriate empiric antimicrobial therapy is associated with increased mortality, including bacterial and fungal etiologies.
   a. Empiric antimicrobial therapy should minimally include an agent active against methicillin-resistant coagulase-negative (e.g., methicillin-resistant *S. epidermidis* [MRSE]) or coagulase-positive (e.g., MRSA) staphylococci. Vancomycin or daptomycin are preferred; linezolid should not be used for empiric management of CLABSI.
   b. Pathogen-specific risk factors, documented colonization, and previous antimicrobial exposure should be considered when choosing empiric antimicrobial therapy.
      i. Patients at risk of MDROs should receive combination therapy against gram-negative bacilli using agents from separate antibiotic classes.
      ii. Use of an echinocandin (e.g., anidulafungin, caspofungin, or micafungin) should be considered for patients at risk of candidemia. Fluconazole is reasonable in patients without recent exposure to and low prevalence of nonsusceptible species.
   c. Local antimicrobial activity should be considered to increase the probability of appropriate therapy.
   d. Empiric antimicrobial therapy should be de-escalated, depending on the identified pathogen(s) and related antimicrobial susceptibility. Antimicrobial therapy should be discontinued if a CLABSI is not evident and there are no other sources of infection.
4. Definitive management and antimicrobial therapy should be based on whether the CLABSI is complicated or uncomplicated (more common than complicated) and the identified pathogen(s). Ongoing clinical trials may provide guidance to the optimal duration of antibiotic therapy for CLABSI and non–central line-associated bacteremia (BMJ Open 2020;10:e038300).

a. The duration of antimicrobial therapy should be based on the first day of negative blood culture.
b. Complicated CLABSI
    i. Endocarditis; immunosuppression (*S. aureus* only); diabetes (*S. aureus* only); chronic intravascular hardware; osteomyelitis; positive blood cultures greater than 72 hours from initiation of appropriate therapy; septic thrombus; thrombophlebitis
    ii. Remove catheter.
    iii. Treat with pathogen-targeted antimicrobial therapy for 4–6 weeks; 6–8 weeks for osteomyelitis.
c. Uncomplicated CLABSI
    i. Coagulase-negative staphylococci
        (a) Consider catheter removal. If catheter is retained, consider antibiotic lock therapy in addition to systemic antibiotic therapy for 10–14 days.
        (b) Treat with systemic antibiotic therapy for 5–7 days if the catheter is removed.
    ii. *S. aureus*
        (a) Remove catheter.
        (b) Treat with systemic antibiotic therapy for a minimum of 14 days.
            (1) Methicillin-sensitive *S. aureus* (MSSA) – Penicillinase-resistant penicillin (e.g., nafcillin); first-generation cephalosporin (e.g., cefazolin) (Note: Vancomycin has been shown inferior to β-lactam therapy for MSSA.)
            (2) MRSA – Vancomycin; daptomycin or linezolid; sulfamethoxazole/trimethoprim
        (c) Patients with catheter tip bacterial growth but negative blood cultures should receive antibiotic therapy for 5–7 days with close monitoring for signs and symptoms of ongoing infection and consideration for repeat blood cultures.
    iii. *Enterococcus* spp.
        (a) Remove catheter.
        (b) Treat with systemic antibiotic therapy for 7–14 days.
    iv. Gram-negative bacilli
        (a) Remove catheter.
        (b) Treat with systemic antibiotic therapy for 7–14 days.
    v. *Candida* spp.
        (a) Remove catheter.
        (b) Treat with systemic antifungal therapy for at least 14 days.
    vi. Antibiotic lock therapy in the treatment of CLABSI should be used primarily for catheter salvage. In the event antibiotic therapy cannot be used in this situation, then recommendation is to administer antibiotics through the colonized catheter.

Patient Cases

4. T.W. is a 47-year-old woman admitted to the MICU with respiratory failure secondary to severe 2009 H1N1 influenza infection. T.W., who requires intubation and mechanical ventilation, is given a diagnosis of septic shock associated with influenza and a secondary MSSA pneumonia. An internal jugular vein CVC was placed in the ED during acute resuscitation. T.W. continues to require a CVC. Although her hypotension and fever resolved 72 hours post-admission, she has a new temperature of 101.7°F (38.7°C) with worsening leukocytosis on ICU day 5; there is no change on her chest radiograph. Which action would be best to take next?

    A. Initiate broad-spectrum antibiotic therapy for a new sepsis episode.
    B. Perform bronchoscopic BAL for suspected VAP.
    C. Remove CVC.
    D. Send two sets of blood cultures, one from the catheter and one from a peripheral blood sample.

Patient Cases *(continued)*

Questions 5 and 6 pertain to the following case.
F.P. is a 29-year-old man admitted to the MICU from another hospital with severe, alcohol-induced, sterile acute pancreatitis. During his 5-day stay at the outside hospital, he had multiple organ failure, including respiratory failure requiring tracheostomy. A right internal jugular CVC was placed on the patient's admission to the outside hospital. On MICU day 3, F.P. has a maximum temperature of 102°F (38.9°C). Two sets of blood cultures are obtained; the right internal jugular catheter is removed and the tip sent for culture.

5. Which regimen would be best to consider for empiric management of a suspected CLABSI?
   A. Daptomycin and ceftriaxone.
   B. Fluconazole.
   C. Linezolid and cefepime.
   D. Vancomycin, cefepime, and tobramycin.

6. F.P. is found to have *K. pneumoniae* CLABSI and receives appropriate empiric antibiotic therapy. He has had a good clinical response with no persistence of bacteremia. Which represents the best duration of F.P.'s definitive antibiotic therapy for CLABSI?
   A. 5 days.
   B. 14 days.
   C. 21 days.
   D. 4 weeks.

III. CATHETER-ASSOCIATED URINARY TRACT INFECTIONS

A. Epidemiology
   1. Urinary catheters are commonly used in critically ill patients, with documented use of 50%–80% of adult critically ill patient-days. Urinary catheters most often used are short term (less than 30 days) temporary indwelling or intermittent urethral catheters. Other catheters include suprapubic catheters or external collection catheters (e.g., condom catheters). Presence of an indwelling urinary catheter is an independent risk factor for urinary tract infection (UTI).
   2. CAUTIs are the most common cause of infection in critically ill patients. In 2012, the incidence (cases per 1000 catheter-days) of CAUTIs varied among acute care critically ill populations and ranged from 1.2 for medical/surgical patients to 4.7 for burn patients. Aggregated data from 2016 show an almost 50% decrease in CAUTI incidence since 2012, likely related to organized prevention programs and implementation of best practices for catheter use.
   3. CAUTIs account for 15%–21% of hospital-acquired bacteremias.
   4. The attributable mortality for CAUTI is 0%–15%.

B. Definitions
   1. CDC adult surveillance definitions
      a. UTI – At least one of the following signs or symptoms: Fever (temperature greater than 100.4°F [38°C]); suprapubic tenderness; or costovertebral angle pain or tenderness and one of the following:

i. A positive urine culture of 10^5 CFU/mL or more of no more than two species of microorganisms
ii. At least one of the following findings: Positive dipstick for leukocyte esterase and/or nitrite, pyuria (urine specimen with at least 10 white blood cells/mm^3 of unspun urine or greater than 5 white blood cells/high-power field of spun urine), or microorganisms seen on Gram stain of unspun urine and a positive urine culture of 10^3–10^5 CFU/mL of no more than two species of microorganisms

b. CAUTI – A UTI in which an indwelling urinary catheter was in place for greater than 2 days and attributable to the catheter removed no more than 1 day before infection

2. IDSA adult definitions
   a. Catheter-associated asymptomatic bacteruria – Patients with indwelling urethral, indwelling suprapubic, or intermittent catheterization (or condom catheter in a man) is defined by the presence of 10^5 CFU/mL or greater of one or more bacterial species in a single catheter urine specimen in a patient without symptoms compatible with UTI
   b. CAUTI – In patients with indwelling urethral, indwelling suprapubic, or intermittent catheterization, CAUTI is defined by the presence of symptoms or signs compatible with a UTI with no other identified source of infection, together with 10^3 CFU/mL or more of one bacterial species in a single catheter urine specimen or in a midstream-voided urine specimen from a patient whose urethral, suprapubic, or condom catheter has been removed within the previous 48 hours. Lower colony counts are more likely to represent significant bacteriuria in a symptomatic person than in an asymptomatic person.

C. Etiology
1. Most pathogens causing CAUTIs are acquired from the external environment, including the urethra, the catheter collection system, and local skin flora. Most short-term catheter CAUTIs are monomicrobial. Longer duration of an indwelling catheter is associated with the formation of biofilms within the catheter and related system, which can promote polymicrobial infections.
2. *E. coli* is the most prevalent pathogen causing CAUTIs; however, because of a wider distribution of pathogens compared with non–catheter-associated UTIs, *E. coli* accounts for only about one-third of CAUTIs. Additional bacterial pathogens include other enteric gram-negative bacilli (e.g., *Klebsiella* spp.; *Proteus* spp.; *Enterobacter* spp.), non–lactose-fermenting gram-negative bacilli (e.g., *P. aeruginosa*), and gram-positive cocci (e.g., *Enterococcus* spp.; MSSA; MRSA; MRSE). *Candida* spp. may be involved in up to one-third of CAUTIs.
3. Similar to other health care–associated infections, causative pathogens are associated with local pathogen patterns, pathogen-specific risk factors, and patient severity of illness.

D. Prevention
1. CAUTI is considered a preventable complication. The NHSN, using CDC definitions, provides population-specific event rates for institution surveillance and performance benchmarks.
2. The most effective way to reduce the incidence of catheter-associated asymptomatic bacteruria and catheter-associated UTI is to reduce the use of urinary catheterization by restricting its use to patients who have clear indications and by removing the catheter as soon as it is no longer needed.
3. The foundation for preventing CAUTIs includes training and education, proper aseptic insertion techniques, and active surveillance and performance improvement systems.
4. Recommended best practices for preventing CAUTIs have been proposed and endorsed by IDSA, SHEA, and the Joint Commission. These evidence-based recommendations are categorized as essential practices for all acute care hospitals or special practices wherein basic practices are less than effective at reducing CAUTI rates. The entire document is available (Infect Control Hosp Epidemiol 2023;44:1209-31). Major recommendations include:

a. Essential practices
   i. Insert urinary catheters only when necessary for patient care and leave in place only as long as indications remain.
   ii. Provide comprehensive education to and ensure competency for all involved with insertion, care, and maintenance of urinary catheters.
   iii. Use of a systematic process or checklist at the time of insertion to ensure adherence to proper insertion technique
   iv. Practice hand hygiene before insertion of the catheter, as well as before and after any manipulation of the catheter site or apparatus.
   v. Proper care and maintenance of indwelling catheter and collection system
b. Additional approaches
   i. Establish a system for analyzing and reporting data on catheter use.
   ii. Development and implementation of a protocol to manage postoperative urinary retention

E. Diagnosis
1. Although clinical signs and symptoms are the mainstay for differentiating CAUTIs from catheter-associated asymptomatic bacteruria, they are not specific for CAUTI. These include fever, rigors, altered mental status, malaise, or lethargy with no other identified cause. Presence of flank pain, costovertebral angle tenderness, acute hematuria, and pelvic discomfort may be more specific, but these are difficult to assess in many critically ill patients.
2. Urine culture should be obtained in all critically ill patients with signs and symptoms of CAUTI. Sampling should be done through the catheter port using aseptic technique. Catheters in place for longer than 2 weeks should be replaced and a urine sample obtained from the port of the newly placed catheter. Samples from the catheter collection system (e.g., catheter bag) should be avoided because of the potential for colonization.
3. Urinalysis should be obtained in patients with a suspected CAUTI. Pyuria without signs and symptoms does not indicate CAUTI; however, the absence of pyuria in a symptomatic patient suggests a diagnosis other than CAUTI.

F. Treatment
1. A urinary culture should be obtained before initiation of antimicrobial therapy.
2. If an indwelling catheter has been in place for longer than 2 weeks, the catheter should be removed or, if still indicated, replaced to hasten resolution of symptoms.
3. Similar to other health care–acquired infections, empiric antimicrobial therapy should be based on local pathogen prevalence, pathogen-specific risk factors (e.g., MDRO risk factors), previously identified pathogens, previous antimicrobial exposure, and local antibiotic susceptibility.
4. Empiric antimicrobial therapy should be de-escalated according to the identified pathogen(s) and respective antimicrobial susceptibility on final urine culture results. Empiric antimicrobial therapy should be discontinued in patients without CAUTI or other sources of infection.
5. Definitive antimicrobial therapy should be based on final antimicrobial susceptibility results and presence of concomitant infection(s). Catheter irrigation (i.e., bladder washings) is not recommended.
6. Shorter duration of antimicrobial therapy should be considered depending on clinical response. Regardless of catheter removal, duration of antimicrobial therapy should be 7 days for patients with resolution of signs and symptoms within 72 hours of appropriate antimicrobial therapy and up to 14 days in patients with resolution after 72 hours.
7. Patients with persistent signs and symptoms of CAUTI should receive a urologic workup to assess for abscess or other causes of relapse.

Patient Case

Questions 7 and 8 pertain to the following case.
P.H. is a 67-year-old woman admitted from a long-term care facility to the MICU for respiratory failure secondary to severe pneumonia. P.H. is initiated on vancomycin and levofloxacin and has a urethral catheter placed in the emergency department. On ICU day 6, P.H. has a new fever with a temperature of 101.9°F (38.9°C) and an elevated WBC to 18 x 10^3 cells/mm^3, despite an initial clinical response to pneumonia therapy. Blood and urine cultures are sent. Urinalysis reveals many bacteria and greater than 10 white blood cells/mm3.

7. Which intervention is most appropriate in P.H.?
   A. Await final culture results before changing the current antibiotic regimen.
   B. Continue pneumonia therapy for an extended treatment duration.
   C. Initiate empiric antibiotic therapy for a suspected catheter-associated urinary tract infection (CAUTI).
   D. Insert a new urinary catheter and resend a urinalysis to confirm CAUTI.

8. Given P.H.'s risk factors for multidrug-resistant pathogens, which listing of pathogen would most likely be associated with a CAUTI?
   A. *E. coli, P. aeruginosa*, *C. albicans.*
   B. *Enterococcus* spp., *C. albicans*, *S. maltophilia*.
   C. MRSA, *Enterococcus* spp., *C. albicans.*
   D. *P. aeruginosa*, *Enterococcus* spp., MRSA.

IV. COMPLICATED INTRA-ABDOMINAL INFECTION

A. Epidemiology
   1. Complicated intra-abdominal infection spans prehospital and in-hospital dispositions and a variety of pathogenic processes involving several organ systems.
   2. The incidence of complicated intra-abdominal infection is difficult to estimate, given the breadth of illness. Appendicitis is the most prevalent cause of complicated intra-abdominal infection. Recent controlled trials and large international observational sepsis studies report abdominal infection as the second or third most common source of sepsis, accounting for 21% of sepsis and 30% of septic shock cases.
   3. Complicated intra-abdominal infection is the second most common cause of mortality in critically ill patients. Crude mortality for primary intra-abdominal infection is almost 30%, whereas rates exceed 50% in patients with secondary intra-abdominal infection. Those presenting with sepsis caused by intra-abdominal infection have crude mortality above 40%.

B. Definitions
   1. The IDSA and Surgical Infection Society jointly published guidelines for the management of intra-abdominal infection in 2010. In 2017, the Surgical Infection Society updated these guidelines, including management for complicated intra-abdominal infection, which are reflected throughout this section. As an additional resource, an international group published recommendations in 2021 on a global approach to diagnosis and management of specific intra-abdominal infections (World J Emerg Surg 2021;16:49).

2. A complicated intra-abdominal infection is defined as infection that extends beyond the hollow viscus of origin into the peritoneal space and is associated with either abscess formation or peritonitis. These infections usually arise from spillage of viscus-related fluid and flora into the peritoneal cavity, causing inflammation and injury to the peritoneal membrane. Retroperitoneal infections also are possible, but these are related to the individual retroperitoneal organ rather than the peritoneum.
3. Peritonitis is described as primary, secondary, and tertiary.
    a. Primary peritonitis, or spontaneous bacterial peritonitis, is peritonitis related to bacterial translocation of proximal small bowel overgrowth and not peritoneal disruption or organ perforation. Primary peritonitis is generally diffuse in nature.
    b. Secondary peritonitis is caused by leakage of intraluminal fluid and microorganisms secondary to macro- or microperforation of the GI tract. Secondary peritonitis can be diffuse or localized to an organ, depending on the extent of peritoneal involvement. Causes of secondary peritonitis include direct trauma, ischemia, thrombosis, ulceration, malignancy, and anastomotic leak.
    c. Tertiary peritonitis is peritonitis that persists or recurs at least 48 hours after the management of primary or secondary peritonitis. Tertiary peritonitis can represent a new intra-abdominal process or host, anatomic, or therapeutic failure of treatment of primary or secondary peritonitis.

C. Etiology
1. Pathogens associated with complicated intra-abdominal infections and peritonitis are influenced by type/cause of peritonitis, MDRO risk factors, previous antibiotic exposure, and patient-specific colonization. Patients having health care–associated intra-abdominal infections more often tend to have antibiotic-resistant, nosocomial pathogens compared with patients having community-acquired infection.
2. Primary peritonitis also known as spontaneous bacterial peritonitis, is usually monomicrobial and associated with translocation of organisms across the diaphragm or proximal small bowel. The most prevalent organisms include *S. pneumoniae*, *E. coli*, and *Klebsiella* spp.
3. Secondary peritonitis is typically polymicrobial related to the origin of GI tract leakage.
    a. Gastric and duodenal secretions are usually sterile or with limited inocula of gram-positive bacteria and *Candida* spp.
    b. Proximal small bowel is densely populated with aerobic gram-negative bacilli including *E. coli*, *Klebsiella* spp., *Proteus* spp., and Enterobacter spp., as well as populations of aerobic gram-positive bacteria such as *S. aureus*, streptococci, and enterococci.
    c. The distal small bowel and the large bowel are populated with proximal small bowel flora in addition to anaerobic gram-negative and gram-positive organisms, including *Bacteroides fragilis* and *Clostridioides* spp. (usually non–*C. difficile*).
4. Tertiary peritonitis includes core organisms of primary and secondary peritonitis further compounded by the influence of management strategies, including malnutrition, anatomic disruption, and antimicrobial therapy.

D. Diagnosis
1. The diagnosis of complicated intra-abdominal infection relies on assessment of clinical signs and symptoms, differentiation from other causes of infection, and radiographic evaluation.
2. Rapid-onset abdominal pain and tenderness with signs of peritoneal irritation on physical examination are the most common presenting signs and symptoms. Additional symptoms of anorexia, abdominal distention, nausea, or vomiting, with or without fever, tachycardia, or tachypnea. These are often difficult to assess in critically ill patients; therefore, intra-abdominal infection should be considered in patients with unexplained new-onset organ dysfunction and sepsis.

3. Radiographic evaluation is commonly performed using abdominal radiographs, ultrasonography, and CT, including contrast CT to assess for vascular thrombosis. Contrast studies of postoperative drains or fistulae may also help assess anastomotic integrity.
4. Culture of intra-abdominal fluid (e.g., ascitic fluid in patients with cirrhosis; intraoperative culture of abscess fluid) associated with the primary source should be obtained in moderately to severely ill patients. Gram stain results should be used to help guide empiric antimicrobial therapy. Pathogens identified on final culture should guide definitive antimicrobial therapy.

E. Management and Treatment
1. Resuscitation for hemodynamic instability should be initiated and managed according to the Surviving Sepsis Campaign guidelines.
2. Achieving primary source control of ongoing peritoneal contamination by operative diversion or resection is recommended for patients with diffuse peritonitis. Patients with focal peritonitis should undergo percutaneous abscess and fluid drainage.
3. Empiric antimicrobial therapy should be initiated in patients thought to have complicated intra-abdominal infections.
    a. Empiric antimicrobial regimens should be based on the source/location of intra-abdominal infection, community-acquired or health care–associated disposition of infection, severity of infection, local pathogen trends and susceptibilities, MDRO risk factors (e.g., hospitalization for greater than 48 hours during current admission or in the previous 90 days; recent broad-spectrum antimicrobial therapy; infection developing greater than 48 hours after initial source control; home wound care or dialysis within preceding 90 days), and patient-specific colonization patterns.
    b. Individual antimicrobial agents should be dosed according to available pharmacokinetic and pharmacodynamic principles to optimize efficacy and limit toxicity.
    c. Community-acquired, mild to moderate severity, low risk for treatment failure
        i. Routine culture is not recommended.
        ii. Empiric antibiotic therapy should include agents active against aerobic and facultative enteric gram-negative bacilli and enteric streptococci. Obligate anaerobic therapy should be initiated in patients with distal small bowel, appendiceal, or colonic sources. Concerns with emergence of antibiotic resistance may limit the utility of narrower-spectrum agents (e.g., cefazolin; cefoxitin). Anti-enterococcal and antipseudomonal therapies are not recommended.
            (a) Either ceftriaxone or cefotaxime in combination with metronidazole (generally considered first line)
            (b) Cefuroxime plus metronidazole
            (c) Ertapenem
            (d) Moxifloxacin alone; levofloxacin or ciprofloxacin plus metronidazole for patients with severe β-lactam allergies
            (e) Tigecycline
                (1) No longer guideline recommended for empiric use, but can be considered for definitive treatment.
                (2) Pooled results from approval trials across all infections suggests increased mortality in patients treated with tigecycline versus active comparator regimens. The cause of this increase has not been established.
                (3) FDA-approved label warns that the increase in all-cause mortality should be considered when selecting among treatment options.

d. Community acquired, high risk for treatment failure defined as two or more of the following risk factors: Severe physiologic disturbance (e.g., septic shock); advanced age; immunocompromised state; diffuse peritonitis; delay in, or high likelihood of failure to achieve primary source control
   i. Routine culture of intra-abdominal fluid is recommended as available.
   ii. Empiric antimicrobial therapy should be broadened to include MDROs, enterococci, and obligate anaerobes. Agents active against MRSA are not recommended empirically. Although fluoroquinolones are recommended, emergence of resistant *E. coli* and *P. aeruginosa* is a concern.
      (a) Broad-spectrum carbapenem (e.g., imipenem, meropenem)
      (b) Either cefepime or ceftazidime in combination with metronidazole (may need additional anti-enterococcal agent)
      (c) Either ciprofloxacin or levofloxacin in combination with metronidazole (may need additional anti-enterococcal agent)
      (d) Piperacillin/tazobactam (or cefepime/ceftazidime) would not be appropriate if an ESBL infection

e. Health care associated
   i. Routine culture of intra-abdominal fluid is recommended as available.
   ii. Broad-spectrum empiric antimicrobial therapy should include coverage for MDR Gram negative bacilli (e.g., *P. aeruginosa*) and obligate anaerobes. Depending on empiric antibiotic susceptibility rates (i.e., local antibiogram), combination therapy against aerobic gram-negative bacilli with aminoglycoside, or, if MDROs are prevalent, ceftolozane-tazobactam and/or polymyxin.
      (a) Broad-spectrum carbapenem (e.g., imipenem, meropenem with or without anti-enterococcal therapy)
      (b) Either cefepime or ceftazidime (second line agent) in combination with metronidazole.
      (c) Piperacillin/tazobactam
   iii. Anti-enterococcal therapy should be considered in patients with recent exposure to broad-spectrum antimicrobial therapy, septic shock, postoperative infections, and those documented colonization with enterococci.
   iv. Anti-MRSA therapy should be considered in patients known to be colonized with MRSA or with MDRO risk factors, including advanced age, co-morbid medical conditions, previous hospitalization or surgery, and significant recent exposure to antibiotic agents.
   v. Antifungal therapy should be added to the regimen of patients with yeast on Gram stain, recent evidence of heavy colonization, surgically treated pancreatitis, prolonged broad-spectrum antibiotic therapy, or critically ill patients with an upper gastrointestinal source. Fluconazole is the drug of choice for fluconazole-susceptible strains. Echinocandins should be used first line in critically ill patients until final culture results are available.
   vi. Empiric antimicrobial therapy should be de-escalated to final culture results and related antimicrobial susceptibilities.

4. Definitive antimicrobial therapy for complicated intra-abdominal infection should be continued for no more than 4 days in patients who have adequate source control (N Engl J Med 2015;372:1996-2005). Duration of antimicrobial therapy in patients without definitive source control should be no more than 5-7 days, depending on clinical response. Patients with delayed or incomplete clinical response to antimicrobial therapy should be reassessed for additional source control intervention.
5. Considerations for short-term prophylactic courses no longer than 24 hours include:
   a. Acute gastric or jejunal perforation in absence of malignancy or acid-reducing pharmacotherapy with adequate source control within 24 hours
   b. Traumatic or iatrogenic bowel injuries repaired within 12 hours of injury
   c. Acute appendicitis without perforation or abscess

Patient Cases

9. D.L. is a 69-year-old woman admitted from home to the SICU with severe acute abdominal pain in the left lower quadrant. An abdominal CT reveals free air in the peritoneal cavity and evidence of distal ischemic colitis. D.L. is taken urgently to the operating room, where she is noted to have gross contamination from distal colonic perforation with marked peritonitis. After a partial colectomy and abdominal washout, she is admitted to the SICU for continued resuscitation for septic shock. Which empiric antibiotic regimen would be most appropriate for D.L.?
   A. Ceftriaxone and vancomycin.
   B. Ciprofloxacin and metronidazole.
   C. Ertapenem.
   D. Piperacillin/tazobactam.

10. K.D. is a 58-year-old man with severe peripheral vascular disease admitted from home to the SICU with severe acute abdominal pain in the left lower quadrant. An emergency abdominal CT reveals free air in the peritoneal cavity and evidence of distal small bowel ischemia. K.D. is taken emergently to the operating room for an exploratory laparotomy, where he is noted to have gross peritoneal contamination with marked peritonitis from a small bowel perforation. Intraoperative cultures were sent and are pending. After primary repair of the distal small bowel and abdominal washout, K.D. returns to the SICU for continued resuscitation for septic shock. Which empiric antibiotic regimen would be most appropriate for K.D.?
   A. Piperacillin/tazobactam.
   B. Ertapenem.
   C. Ciprofloxacin and vancomycin.
   D. Cefazolin, metronidazole, and fluconazole.

V. ACUTE PANCREATITIS

A. Epidemiology
   1. Acute pancreatitis is responsible for more than 200,000 acute care admissions annually in the United States.
   2. About 20% –30% of acute pancreatitis episodes are severe, have evidence of pancreatic necrosis, and are associated with local and systemic complications (e.g., multiple organ failure). An Acute Physiology and Chronic Health Evaluation II (APACHE II) score of 8 or higher and a Ranson's criteria score of 3 or greater, together with clinical presentation, have been used historically to categorize patients as having severe acute pancreatitis. The more recent revised Atlanta classification categorizes severe acute pancreatitis as the presence of single or multiple organ failure for 48 hours or more, whereas the Determinant-Based Criteria (DBC) classification system defines it as persistent organ failure or infected pancreatic or peripancreatic necrosis. The DBC adds a category of critical acute pancreatitis for patients with persistent organ failure and infected necrosis.
   3. Between 30% and 78% of patients with necrotizing pancreatitis will have concomitant organ failure, with the highest incidence in patients having infected necrosis. Pancreatitis-associated systemic sequelae and organ failure includes:
      a. Systemic inflammatory response syndrome (SIRS)
      b. Hypovolemic shock

   c. Hypoxemia secondary to ARDS
   d. Acute kidney injury
   e. Gastrointestinal (GI) bleeding
   f. Disseminated intravascular coagulation and severe metabolic disturbances can also occur.
4. Although most pancreatic necrosis is sterile and does not require treatment with antimicrobials, about one-third of patients with necrotizing pancreatitis will have infected necrosis.
5. Overall, crude mortality for acute pancreatitis is 2%–9%; however, mortality rates for necrotizing and infected necrotic pancreatitis with organ failure are as high as 44% and 62%, respectively.

B. Pathophysiology
1. Pancreatitis is a result of glandular autodigestion from excessive ductal and tissue exposure to amylase, lipase, and protease caused by trypsin-related hyperstimulation, macro-ductal blockade, or micro-ductal blockade.
2. The pathophysiology of acute pancreatitis can be organized into three phases:
   a. Excessive activation or decreased inactivation of trypsin leading to activation of pancreatic exocrine enzymes
   b. Local inflammatory and immune response to pancreatic injury
   c. Systemic inflammatory and immune response, including SIRS, hypovolemia, and ARDS
3. Common causes of acute pancreatitis include biliary obstruction (e.g., gallstones), direct toxicity (e.g., alcohol), trauma, surgery/biliary procedures, and drugs.

C. Definitions
1. Severe pancreatitis is defined as pancreatitis associated with organ failure that persists for more than 48 hours. Hypovolemia may increase the risk of pancreatic necrosis and intestinal ischemia because of tissue hypoperfusion.
2. Pancreatic necrosis is evidenced on CT scan as diffuse or focal areas of nonviable pancreatic tissue often associated with peripancreatic fat necrosis. In general, greater than 30% of the pancreas should be affected. Infected necrosis is defined as the presence of pathogenic microorganisms in the necrotic tissue. The presence of retroperitoneal gas on CT scan may indicate infected pancreatitis in patients with severe acute pancreatitis. The timing of infected necrosis varies, but infected necrosis usually peaks 2–4 weeks from the onset of acute pancreatitis.
3. Pancreatic pseudocyst is a non-epithelialized wall containing pancreatic excretions caused by acute or chronic pancreatitis or pancreatic trauma. Pseudocysts are usually sterile.
4. Pancreatic abscess is an infected pseudocyst or liquefaction of pancreatic necrosis that becomes infected.
5. Additional late complications of peripancreatic necrosis include secondary pancreatic infection of previously acute necrotic collections during the first 4 weeks after presentation and walled-off necrosis, which may lead to pancreatic abscess that matures after 4 weeks or more.

D. Diagnosis
1. The diagnosis of acute pancreatitis requires two of the following three features:
   a. Acute and constant epigastric or right upper quadrant abdominal pain with or without nausea and vomiting;
   b. Serum amylase and/or lipase greater than 3 times the upper limit of normal; and
   c. Characteristic findings of acute pancreatitis on CT scan or magnetic resonance imaging.
2. Patients with severe pancreatitis may present with SIRS, hypovolemia, and new-onset organ failure, including hypotension.

E. Management and Treatment
   1. Management of severe acute pancreatitis includes acute pain management, fluid resuscitation, supportive care of systemic complications and organ failure, and nutrition support. Surgical debridement of infected necrosis or drainage of pancreatic abscess should be considered.
      a. Initial fluid management should be goal-directed titration of isotonic, crystalloid intravenous fluids using clinical and biochemical targets of perfusion. Results of the ERICA Trial support moderate fluid volumes over more aggressive resuscitation (N Engl J Med 2022;387:989-1000).
      b. Patients should receive nutrition within 24 hours. Patients unable to eat orally should receive enteral nutrition by nasogastric or nasojejunal access over parenteral nutrition.
      c. Patients should be monitored and approriately treated for acute pain during acute pancreatitis. Refer to chapter on management of pain in the ICU.
   2. Antibiotic therapy for acute pancreatitis is considered empiric, definitive, or prophylactic.
      a. Empiric antibiotic therapy is indicated in patients with suspected or confirmed infected necrosis or pancreatic abscess. The diagnosis of infected pancreatitis is difficult because the clinical picture may mimic other infectious complications because of the likelihood of systemic signs of inflammation.
         i. Patients with suspected infected necrosis or pancreatic abscess should undergo radiographically (e.g., ultrasonography, CT) guided drainage with culture of recovered material/fluid. Common organisms associated with infected necrosis or pancreatic abscess include:
            (a) *E. coli*
            (b) *Klebsiella* spp.
            (c) *Enterobacter* spp.
            (d) *Proteus* spp.
            (e) Streptococci
         ii. The antibiotics of choice include:
            (a) Piperacillin-tazobactam or broad-spectrum carbapenems (e.g., imipenem; meropenem)
            (b) Third-or-fourth generation cephalosporin plus metronidazole (limit use to patients with serious β-lactam allergies)
            (c) Fluoroquinolone plus metronidazole
         iii. Antifungal therapy should be added to antibiotic therapy if yeast is identified on culture Gram stain. Fluconazole is recommended unless there is a high suspicion for fluconazole-resistant fungi.
         iv. Empiric antibiotics should be discontinued if pancreatic culture is negative, indicating sterile pancreatitis. Patients with persistent SIRS should undergo repeat imaging and drainage of identified abscess or fluid.
         v. If infection is confirmed, antibiotic therapy should be de-escalated toward the identified pathogen(s) according to antibiotic susceptibility.
         vi. Definitive antibiotic therapy for confirmed infection should be continued for up to 14 days.
      b. Prophylactic antimicrobial therapy in patients with sterile necrotizing pancreatitis is controversial. Although small pilot investigations suggest benefit, a large, noninferiority, placebo-controlled trial suggested no benefit with prophylactic meropenem. Current recommendations do not support routine use of prophylactic antibiotic or antifungal therapy in patients with sterile necrotizing pancreatitis. (Gastroenterology 2018;154:1096-101; Am J Gastroenterol 2013;108:1400-15; World J Emerg Surg 2019;14:27)
   3. Ongoing assessment of resolution of SIRS, trending of procalcitonin, and pancreatitis-associated organ failure is imperative. Serial assessment of serum amylase or lipase has limited value over clinical assessment and physical examination. Elevation of serum amylase or lipase for several weeks, however, should heighten concern for persistent pancreatic/peripancreatic inflammation, blockage of the pancreatic duct, or development of a pseudocyst.

Patient Cases

11. R.P. is a 43-year-old man with a history of chronic alcohol abuse who presents to the emergency department from home with acute, severe epigastric abdominal pain; heart rate 142 beats/minute; MAP 62 mm Hg; WBC 24 x 10^3 cells/mm^3; and serum lipase 1527 U/L. R.P. receives 3 L of normal saline intravenously over 60 minutes. An immediate abdominal CT scan is remarkable for significant pancreatic edema and greater than 30% necrosis of the pancreas. There is no evidence of an acute fluid collection. Which antimicrobial regimen would be most appropriate for R.P.?
    A. Ceftriaxone and vancomycin.
    B. Ciprofloxacin and metronidazole.
    C. Meropenem.
    D. None; prophylactic antibiotic therapy is not indicated in acute pancreatitis.

12. T.J. is a 27-year-old man who is admitted to the MICU from an outside hospital with alcohol-induced acute pancreatitis and associated respiratory failure, SIRS, and blood pressure 100/55 mm Hg. Admission abdominal CT shows severe pancreatitis with about 30% necrosis and a large focal fluid collection requiring guided drainage. A fluid sample is sent for a Gram stain, which reveals many gram-negative rods and yeast. Which antimicrobial regimen would best be initiated in T.J.?
    A. None – low suspicion for infected pancreatitis.
    B. Ciprofloxacin and metronidazole.
    C. Imipenem.
    D. Meropenem and fluconazole.

VI. *CLOSTRIDIOIDES DIFFICILE* INFECTION

A. Epidemiology
    1. *C. difficile* (also known as *Clostridioides difficile*) is a spore-forming, anaerobic, gram-positive bacillus. Stool carriage of *C. difficile* has been found in up to 26% of acute care patients.
    2. *C. difficile* is transmitted person to person through the fecal-oral route. The capability to become dormant in spores increases the potential for environmental spread.
    3. *C. difficile* is pathogenic through the production and expression of two primary toxins, *C. difficile* toxin A (TcdA) and *C. difficile* toxin B (TcdB). North America has seen a recent emergence of strains with increased virulence (e.g., BI/NAP1 strain) through increased production of toxins A and B, production of additional toxins, and enhanced sporification.
    4. CDI is responsible for almost 30% of antibiotic-associated diarrhea and is the most common cause of infectious diarrhea in health care settings. The estimated incidence of CDI is 3–10 cases per 10,000 patient-days. Community-acquired CDI has increased in prevalence.
    5. The clinical manifestations of CDI range from asymptomatic carrier to CDI spanning mild or moderate diarrhea to fulminant and sometimes fatal pseudomembranous colitis, toxic megacolon, or colonic perforation. Post-colectomy small bowel enteritis and rectal pouchitis related to CDI have been reported. Additional complications of severe CDI include:
        a. SIRS
        b. Hypovolemia
        c. Electrolyte disturbances

   d. Sepsis and septic shock
   e. Multiple organ failure
6. CDI is associated with medical costs greater than $3 billion per year.
7. Attributable mortality from CDI is estimated to be below 10%. However, crude mortality is as high as 75% in patients presenting with septic shock, colonic perforation, or toxic megacolon. Subtotal colectomy in patients with a severe CDI is associated with an in-hospital mortality rate of up to 42%.

B. Definitions
1. CDI is defined as the presence of symptoms (e.g., diarrhea) and a stool test result positive for *C. difficile* toxin, toxigenic *C. difficile* (e.g., DNA amplification detecting toxin-coding genes), or pseudomembranous colitis on colonoscopic examination. The most recent guidelines for CDI diagnosis and management include those from the American College of Gastroenterology in 2021 (Kelly CR, et al. Am J Gastroenterol 2021;116:1124 -47), IDSA/SHEA guidelines from 2017, and a focused update from IDSA/SHEA in 2021 related to management of CDI. The ACG 2021 guidelines are updated from the 2013 version to harmonize CDI severity categories with IDSA/SHEA while expanding a focus on diagnostic methods and management of patients with concomitant chronic inflammatory bowel disease.
2. Severe CDI according to expert-based guidelines is defined as stated earlier, plus:
   a. IDSA 2017 and 2021; ACG 2021
      i. WBC 15 x 10^3 cells/mm^3 or greater and
      ii. Serum creatinine (SCr) 1.5 times or greater than the premorbid level
3. Fulminant (IDSA 2017 and 2021; ACG 2021) CDI is defined as:
   a. Severe CDI plus one of the following:
      i. Hypotension or evidence of shock
      ii. Ileus
      iii. Megacolon
4. Recurrence is defined as the relapse of a recent infection or a reinfection after definitive therapy.
5. Additional categorizations of CDI are used for infection control surveillance and institutional/ICU quality review. These are related to the locations of *C. difficile* acquisition (e.g., community acquired or health care associated) and onset of symptoms (e.g., community onset vs. health care onset).

C. Risk Factors
1. Antibiotic therapy is the most important risk factor.
   a. All antibiotic classes have been associated with CDI.
   b. Highest-risk (listed by decreasing risk) antibiotic classes include clindamycin, fluoroquinolones, cephalosporins/carbapenems, and penicillins.
2. Gastric acid–suppressing pharmacotherapy, including proton pump inhibitors and histamine-2 receptor antagonists
3. Age older than 65 years
4. Duration of hospitalization
5. Cancer chemotherapy
6. GI surgery
7. Previous CDI

D. Prevention – There are two global strategies to prevent CDI:
1. Decrease the risk of acquiring *C. difficile*.
   a. Staff, patient, family, caregiver education
   b. Hand hygiene to remove *C. difficile* spores through non–alcohol-based handwashing using soap or chlorhexidine gluconate and water

c. Contact isolation, including full-barrier precautions (gown and gloves), single-occupancy room, and the cohorting of patients with a CDI
d. Limit reuse or between-patient sharing, and terminal clean patient care–related equipment (e.g., digital thermometers, point-of-care blood glucose machines, dietary trays, intravenous infusion pumps) and rooms (previous antibiotic use by the previous bed occupant may increase the risk of CDI in the current occupying patient).
e. Environmental decontamination using bleach-containing cleaning solution

2. Avoid or address reversible risk factors.
   a. Limit overuse of, and discontinue unnecessary, antibiotic therapy.
   b. Decrease duration of hospital stay.
   c. Discontinue unnecessary gastric acid–reducing pharmacotherapy.

E. Diagnosis
1. Diagnosis of CDI is based on clinical and laboratory findings.
2. Clinical findings include presence of diarrhea, defined as passage of three or more unformed stools within 24 consecutive hours.
   a. Rarely, a symptomatic patient will present with ileus and colonic distension with minimal or no diarrhea.
   b. Patients with cecal CDI or right-sided CDI colitis may have formed stools.
3. Laboratory findings include stool sample positive for toxigenic *C*. *difficile*, *C*. *difficile* toxin, or colonoscopic or histopathologic findings showing pseudomembranous colitis. Available strategies for detecting toxin-producing *C*. *difficile* include:
   a. Testing for *C*. *difficile* should only be performed on unformed stool unless patients have ileus. Identifying the ideal testing strategy remains difficult. Institution-specific decisions for which test(s) to use should be evidence based and collaborative across interested parties. Moreover, institutions should consider creating local interdisciplinary guidelines for *C*. *difficile* testing to avoid positive results in colonized patients without infection.
   b. Stool culture with detection of a toxigenic isolate through identification of neutralizable toxin activity is considered the gold standard test. However, this process could take up to 9 days, limiting its clinical utility.
   c. Enzyme immunoassay (EIA) tests for *C*. *difficile* toxins A and B are rapid and widely available; however, poor sensitivity may limit their utility. Obtaining serial samples has been used to increase sensitivity. Sensitivity may also be increased using a two-step process with initial EIA to detect the *C*. *difficile* antigen glutamate dehydrogenase and a follow-up stool culture with detection of the toxigenic isolate.
   d. PCR testing is rapid, sensitive, and specific. Widespread availability may be limited.
   e. The nucleic acid amplification test (NAAT) involves an illumigene *C*. *difficile* assay, which uses loop-mediated isothermal DNA amplification to detect a specific genetic region responsible for coding toxins A and B.
4. Recommended laboratory diagnostic pathways differ regarding whether an institution-specific protocol for stool sample quality based on CDI likelihood is predefined. Briefly, the absence of predefined criteria requires a multistep laboratory process that includes a toxin test (e.g., NAAT plus toxin assay) rather than a NAAT alone. In contrast, the presence of predefined criteria for stool quality and clinical presentation suggestive of CDI permits the use of NAAT alone or a multistep process over a strategy using a toxin assay alone.
5. The same diagnostic criteria are used for recurrent CDI.

F. Management and Treatment
   1. Removal of potential cause(s), as appropriate (e.g., discontinuation of associated antibiotic therapy)
   2. Assessment of disease severity (mild or moderate vs. severe) and whether episode is initial or recurrent
   3. Evaluation of need for surgical intervention
      a. Subtotal colectomy with rectal preservation should be considered for severely ill patients.
      b. Alternative colon-sparing operative surgical strategies have been described.
   4. Antibiotic therapy targeted against *C. difficile* (Table 3)
      a. Samples for diagnostic testing for *C. difficile* should be obtained before empiric antibiotic therapy is initiated. Initiation of empiric antibiotic therapy before final test results should be based on clinical assessment.
      b. Antibiotic choices and respective routes of administration should be based on severity of CDI and ability to achieve relevant intraluminal antibiotic concentrations relative to CDI. Special considerations include:
         i. Fidaxomicin, an oral, limited-bioavailability macrolide antibiotic, is indicated for treatment of CDI. The 2021 IDSA/SHEA focused guideline update recommends (conditional recommendation; moderate certainty of evidence) fidaxomicin as first-line therapy for first episode, non-fulminant CDI when access to fidaxomicin is not inhibited by limited resources. This recommendation is supported by systematic review and meta-analysis of four studies noting increased sustained response to therapy at 4 weeks with fidaxomicin compared to oral/enteral vancomycin. Initial cure, all-cause mortality, and drug adverse events were similar between fidaxomicin and vancomycin. Data supporting fidaxomicin use in critically ill patients are limited to small representative sample sizes and retrospective or observational studies.
         ii. Oral or enteral vancomycin continues to be an alternative for an initial episode of CDI when access to and availability of fidaxomicin is limited. The optimal enteral vancomycin dosage in critically ill patients is unknown. Although ACG and IDSA guidelines recommend enteral vancomycin dosages of 125 mg four times daily for uncomplicated/nonfulminant CDI (Table 3), some practices consider dosages of 250 mg four times daily if there is lack of response or if there is concern for subclinical GI dysmotility (e.g., incomplete enteral nutrition tolerance, elevated gastric residuals, abdominal distention) (Int J Antimicrob Agents 2013;42:553-8; Clin Infect Dis 2015;61:934-41).
         iii. Metronidazole is primarily only recommended in addition to oral vancomycin for the treatment of severe-complicated/fulminant CDI. The IDSA guidelines indicate that metronidazole is a valid treatment option for "initial episode, non-severe" CDI when fidaxomicin and oral vancomycin are unavailable or for patients with limited access to resources to attain these medications.
         iv. For enema volumes in patients requiring rectal instillation of vancomycin, location of CDI-affected area(s) and risk of colonic perforation should be considered. Patients receiving vancomycin enemas may need a cuffed rectal delivery device/tube to facilitate retention.
         v. Patients with CDI-related colitis and proximal colonic diversion (i.e., no continuity with oral or gastric route) may require rectal instillation of vancomycin enema.
         vi. Fecal transplantation has improved outcomes compared with oral vancomycin in noncritically ill patients with recurrent CDI. Logistical challenges with product availability or stool sample collection, fecal delivery and administration, and risk of colonization with pathogenic organisms (e.g., E. coli) should be considered.
         vii. There are no definitive recommendations for duration of CDI antibiotic therapy or prevention of recurrence when non-CDI antibiotic therapy is continued concurrently.
      c. Response to therapy should be assessed by evaluating clinical signs and symptoms, including resolution of diarrhea, laboratory abnormalities, sepsis, and related organ failure. Results of stool testing for *C. difficile* in patients with resolution of disease do not predict recurrence.

Table 3. Treatment Options for *Clostridioides difficile* Infection

Clinical Definition	ACG 2013	IDSA/SHEA 2018	IDSA/SHEA 2021 Focused Update	ACG 2021
Initial episode, nonsevere	Metronidazole 500 mg PO/NG/FT TID for 10 days	Vancomycin 125 mg PO/NG/FT 4 times daily for 10 days —OR— Fidaxomicin 200 mg BID for 10 days *Alternative*: Metronidazole 500 mg PO/NG/FT TID for 10–14 days (delayed response)	Fidaxomicin 200 mg PO/NG/FT BID for 10 days *Alternative:* Vancomycin 125 mg PO/NG/FT 4 times daily for 10 days *Alternative for nonsevere infection if above agents are unavailable*: Metronidazole 500 mg PO/NG/FT TID for 10–14 days (delayed response)	Fidaxomicin 200 mg PO/NG/FT BID for 10 days (strong recommendation; moderate-quality evidence) —OR— Vancomycin 125 mg PO/NG/FT 4 times daily for 10 days (strong recommendation; low-quality evidence)
Initial episode, severe	Vancomycin 125 mg PO/NG/FT 4 times daily for 10 days	Vancomycin 125 mg PO/NG/FT 4 times daily for 10 days —OR— Fidaxomicin 200 mg BID for 10 days		Vancomycin 125 mg PO/NG/FT 4 times daily for 10 days (strong recommendation; low-quality evidence) —OR— Fidaxomicin 200 mg PO/enteral BID for 10 days (conditional recommendation; very low-quality evidence)
Initial episode, severe complicated (ACG)/ fulminant (IDSA)	Vancomycin 125 mg PO/NG/FT 4 times daily + metronidazole 500 mg IV TID *Ileus*: Vancomycin 500 mg PO/NG/FT and per rectum (NS 500 mL) + metronidazole 500 mg IV TID	Vancomycin 500 mg PO/NG/FT 4 times daily + metronidazole 500 mg IV TID *Ileus*: Add rectal vancomycin 500 mg/ NS 100 mL instilled for 1 hr 4 times daily	Vancomycin 500 mg PO/NG/FT 4 times daily + metronidazole 500 mg IV TID *Ileus*: Add rectal vancomycin 500 mg/ NS 100 mL instilled for 1 hr 4 times daily	Same as IDSA/SHEA 2021 Surgery: Suggest total colectomy with end ileostomy and stapled rectal stump or diverting loop ileostomy with colonic lavage and intraluminal vancomycin

Table 3. Treatment Options for *Clostridioides difficile* Infection *(continued)*

Clinical Definition	ACG 2013	IDSA/SHEA 2018	IDSA/SHEA 2021 Focused Update	ACG 2021
First recurrence	Same regimen as used for initial episode, unless severe; if severe, vancomycin should be used	Vancomycin 125 mg 4 times daily for 10 days if metronidazole was used for initial episode —OR— Prolonged tapered and pulsed vancomycin regimen if standard regimen was used for initial episode; for example: 125 mg 4 times daily for 10–14 days, BID for 1 wk, once daily for 1 wk, then every 2 or 3 days for 2–8 wk —OR— Fidaxomicin 200 mg PO/NG/FT BID for 10 days if vancomycin was used for initial episode	Fidaxomicin 200 mg PO/NG/FT BID for 10 days—OR—BID for 5 days, then once every other day for 20 days *Alternative*: Vancomycin PO in tapered and pulsed regimen —OR— Vancomycin 125 mg given 4 times daily PO for 10 days *Adjunctive therapy*: Bezlotoxumab 10 mg/kg IV once during administration of above antibiotics	Recommend fidaxomicin for first recurrence after initial course of vancomycin or metronidazole (conditional recommendation, moderate-quality evidence) Suggest tapering/pulsed dose vancomycin first recurrence after initial course of fidaxomicin, vancomycin, or metronidazole (strong recommendation, very low-quality evidence) Suggest bezlotoxumab for high risk of recurrence (conditional recommendation, moderate-quality evidence)

Table 3. Treatment Options for *Clostridioides difficile* Infection *(continued)*

Clinical Definition	ACG 2013	IDSA/SHEA 2018	IDSA/SHEA 2021 Focused Update	ACG 2021
Second recurrence	Vancomycin PO, then 28-day dosage taper	Vancomycin in tapered and pulsed regimen —OR— Vancomycin 125 mg PO 4 times daily for 10 days, then rifaximin 400 mg TID for 20 days —OR— Fidaxomicin 200 mg PO/NG/FT BID for 10 days —OR— Fecal microbiota transplantation	Fidaxomicin 200 mg PO/NG/FT BID for 10 days—OR—BID for 5 days, then once every other day for 20 days —OR— Vancomycin PO in tapered and pulsed regimen —OR— Vancomycin 125 mg 4 times daily PO for 10 days, then rifaximin 400 mg TID for 20 days —OR— Fecal microbiota transplantation (preferably after 3 antibiotic courses are attempted and completed) <u>*Adjunctive therapy*</u>: Bezlotoxumab 10 mg/kg IV once during administration of above antibiotics	Recommend fecal microbiota transplant (FMT) to prevent further recurrences (strong recommendation, moderate-quality evidence) Suggest bezlotoxumab for high risk of recurrence (conditional recommendation, moderate-quality evidence)

ACG = American College of Gastroenterology; BID = twice daily; FT = feeding tube; IDSA = Infectious Diseases Society of America; IV = intravenously; NS = normal saline; PO = orally; SHEA = Society for Healthcare Epidemiology of America; TID = 3 times daily.

Information from: Johnson S, Lavergne V, Skinner AM, et al. Clinical practice guideline by the Infectious Diseases Society of America (IDSA) and Society for Healthcare Epidemiology of America (SHEA): 2021 focused update guidelines on management of Clostridioides difficile infection in adults. Clin Inf Dis 2021. Available at www.idsociety.org/globalassets/idsa/practice-guidelines/cdi-2021-focused-update.pdf. McDonald LC, Gerding DN, Johnson S, et al. Clinical practice guidelines for Clostridium difficile infection in adults and children: 2017 update by the Infectious Diseases Society of America (IDSA) and Society for Healthcare Epidemiology of America (SHEA). Clin Infect Dis 2018;66:987-94. Surawicz CM, Brandt LJ, Binion DG, et al. Guidelines for Diagnosis, Treatment, and Prevention of Clostridium difficile Infections. Am J Gastroenterol 2013;108:478–98.; Kelly CR, Fischer M, Allegretti JP, et al. ACG Clinical Guidelines: Prevention, Diagnosis, and Treatment of Clostridioides difficile Infections, Am J Gastroenterol 2021; 116: 1124-1147.

Patient Cases

13. K.L. is a 57-year-old man admitted to the MICU with diffuse abdominal pain, temperature 101.9°F (38.8°C), WBC 27 x 10^3 cells/mm^3, heart rate 125 beats/minute, and mean arterial pressure 57 mm Hg. An abdominal CT scan reveals a large amount of intestinal air, suggestive of ileus and moderate transverse and sigmoid colonic inflammation. On review of the patient's history, it is learned that K.L. recently completed a 4-week course of broad-spectrum antibiotic therapy for a postoperative osteomyelitis after repair of right comminuted femur fracture, increasing the suggestion of CDI. Which regimen would be best for empiric management of a suspected CDI in K.L.?
    A. Metronidazole 500 mg intravenously every 8 hours.
    B. Vancomycin 500 mg orally/by nasogastric tube/feeding tube four times daily, metronidazole 500 mg intravenously every 8 hours, and intracolonic vancomycin 500 mg instilled every 8 hours.
    C. Metronidazole 500 mg orally or by nasogastric tube every 8 hours and vancomycin 250 mg orally or by nasogastric tube every 6 hours.
    D. Vancomycin 250 mg orally or by nasogastric tube every 6 hours.

14. E.A. is a 79-year-old man admitted to the MICU for severe pneumonia. On day 5 of treatment with broad-spectrum antibiotic therapy despite negative culture, E.A. is noted to have several liquid bowel movements requiring a rectal pouch (around 1.6 L of stool in past 24 hours), a low-grade fever, elevation of WBC at 27 x 10^3 cells/mm^3, heart rate 124 beats/minute, MAP 67 mm Hg, lactate 1.6 mmol/L, and SCr 1.5 mg/dL. A stool sample is sent for a suspected *C. difficile* infection (CDI). Which regimen would be the best empiric management of a suspected CDI in E.A.?
    A. Metronidazole 500 mg intravenously every 8 hours and intracolonic vancomycin 500 mg instilled every 8 hours.
    B. Metronidazole 500 mg intravenously every 8 hours.
    C. Metronidazole 500 mg orally or by nasogastric tube every 8 hours and vancomycin 125 mg orally or by nasogastric tube every 6 hours.
    D. Vancomycin 125 mg orally or by nasogastric tube every 6 hours.

VII. WOUND INFECTION

A. Epidemiology
    1. Postoperative wound infection is the most common health care–associated infection, affecting up to 5% of inpatient surgery patients. There are 150,000–300,000 cases of postoperative wound infections annually in the United States.
    2. Most postoperative wound infections are mild or moderate in severity. Major complications associated with severe postoperative wound infections include wound dehiscence, reoperation, sepsis, and necrotizing fasciitis. Infections may be classified by severity of illness: Class 1: no systemic signs/symptoms or comorbidities; Class 2: systemic signs/symptoms with stable comorbidities or no systemic signs/symptoms with at-risk comorbidity (e.g., diabetes, obesity, immunosuppression); Class 3: severe systemic signs/symptoms (fever, tachycardia, tachypnoea, and/or transient hypotension); Class 4: sepsis or septic shock, life-threatening infection (e.g., necrotizing fasciitis).

3. Patients with postoperative wound infection have more than a 2-fold higher risk of death compared with those without infection. Crude mortality in patients with streptococcal necrotizing fasciitis and shock (e.g., streptococcal toxic shock) is high, ranging from 30% to 70%.

B. Definitions
   1. Superficial incisional: Infection involving only the skin or subcutaneous tissue of the incision
   2. Deep incisional: Infection involving fascia and/or muscular layers
      a. Deep incision primary: Wound infection in the primary incision in a patient who has had an operation with one or more incisions
      b. Deep incision secondary: Wound infection in a secondary incision in a patient who has had an operation with more than one incision
      c. Necrotizing fasciitis is an aggressive, deep incisional infection tracking along the superficial fascia, which consists of the tissues between the skin and the underlying muscles. Necrotizing fasciitis often results in major tissue destruction. Fournier gangrene is a variant of necrotizing fasciitis involving the scrotum and penis or vulva.
   3. Organ or space: Infection involving any space or organ, opened or manipulated during the procedure, excluding skin incision, fascia, or muscle layers.

C. Etiology
   1. Pathogens causing postoperative wound infections are often related to local flora present on the skin at the time of incision and flora associated with organs/tissues involved in the operative procedure.
   2. Prevalence of drug-resistant strains (e.g., MRSA, multidrug-resistant gram-negative bacilli) depends on local patterns of infection and patient colonization.
      a. The most common bacterial pathogens causing postoperative wound infection are skin flora, including staphylococci and streptococci.
      b. Bacterial pathogens related to anatomic location of the operation:
         i. Upper GI tract (gastric, biliary, proximal small intestine)
            (a) Biliary: Aerobic and anaerobic gram-negative and gram-positive organisms
            (b) Non-biliary: Enteric, aerobic gram-negative bacilli
         ii. Lower GI tract (distal small bowel; colon): Mixed gram-positive and gram-negative flora, facultative and anaerobic
         iii. Female genitalia: Mixed gram-positive and gram-negative flora, facultative and anaerobic
         iv. Axilla: Aerobic gram-negative organisms
         v. Perineum: Aerobic gram-negative and mixed anaerobic organisms
         vi. Respiratory: Aerobic gram-positive and gram-negative organisms
      c. Necrotizing fasciitis
         i. Most infections are monomicrobial, caused predominantly by group A streptococci (*Streptococcus pyogenes)*, *S. aureus*, and anaerobic streptococci (e.g., *Peptostreptococcus*). Non–*C. difficile* clostridia (e.g., *Clostridium septicum*) and *Aeromonas hydrophilia* also are associated with monomicrobial infection. Additional common pathogens in patients presenting with Fournier gangrene include *E. coli* and *Bacteroides* spp.
         ii. Polymicrobial infections usually involve a broad range of pathogens, including aerobic and anaerobic gram-positive and gram-negative organisms. *S. pyogenes* should be empirically suspected in all necrotizing infections. Likelihood of polymicrobial infection is increased if associated with:
            (a) Decubitus ulcers
            (b) Injection sites in illicit drug users
            (c) Intestinal operation

(d) Penetrating abdominal trauma
(e) Perianal abscess
(f) Spread from genitalia (i.e., Fournier gangrene)

D. Prevention
1. Using evidence-based guidelines can prevent up to 60% of postoperative wound infections. Wound infections after elective operation are considered preventable and are reportable health care–associated infections.
2. Many organizations and agencies promote prevention and monitor the prevalence of postoperative wound infection, including the CDC, the Surgical Care Improvement Project (SCIP), the Joint Commission, and the Centers for Medicare & Medicaid Services.
3. Recommended strategies are basic or special approaches.
   a. Essential approaches include:
      i. Administer preoperative, weight-based antibiotic therapy according to operative site and level of expected operative field contamination. Timing of antibiotic therapy should maximize blood and tissue concentrations at the time of incision. Antibiotics should be redosed every 2 half-lives for prolonged procedures. Antibiotic duration should be limited to 24 hours unless there is evidence of peritonitis or active infection during the operative procedure.
      ii. Avoid use of hair removal. If necessary, use clippers or depilatory agent.
      iii. Maintain glycemic control in immediate postoperative period; goal should be to avoid glucose above 180 mg/dL.
      iv. Avoid perioperative hypothermia.
      v. Educate staff, patients, and their families about SSI prevention.
      vi. Perform surveillance for SSI and increase efficiency of surveillance by using automated data.
      vii. Aseptic, intraoperative wound lavage
   b. Additional approaches include:
      i. Preoperative screening for *S. aureus* and consideration of decontamination in patients undergoing orthopedic or cardiothoracic procedures
      ii. Use antiseptic-impregnated sutures as a strategy to prevent SSI.

E. Diagnosis
1. Postoperative wound infections most commonly occur 48 hours after the procedure. Fever in the first 48 hours is usually idiopathic or from noninfectious causes.
2. Wounds should be physically examined serially until healed. Purulent material should be collected aseptically and sent for Gram stain and culture. Cultures in patients with suspected deep tissue infections should be obtained from deep tissues with concomitant blood cultures.
3. Signs and symptoms of superficial incisional postoperative wound infection include:
   a. Purulent incisional drainage
   b. Local pain or tenderness, swelling, and erythema after the incision is opened
   c. Positive culture of purulent drainage
4. Necrotizing fasciitis should be suspected if the following are present:
   a. Severe pain that seems disproportional to the appearance of the wound
   b. Failure to respond to initial antibiotic therapy
   c. Hard, wooden feel of the subcutaneous tissue, often extending beyond the area of affected skin
   d. Crepitus on physical examination or radiographic (radiograph, CT scan) finding, indicating gas in subcutaneous tissues
   e. Skin necrosis or ecchymoses
   f. Sepsis or septic shock

F. Management and Treatment
   1. Opening of the incision, evacuation of the infected material, and continued dressing changes are the foundation of treatment for confirmed postoperative wound infections.
   2. Antibiotic therapy targeted against likely pathogens should be initiated in patients with systemic signs and symptoms or suspected deep tissue infection.
   3. Superficial incisional infection
      a. Erythema and induration less than 5 cm and minimal systemic signs of infection (no fever, no leukocytosis, and tachycardia):
         i. Serial dressing changes
         ii. No antibiotic therapy necessary
         iii. Continue to assess wound for resolution or progression.
      b. Erythema and induration greater than 5 cm, fever, leukocytosis, and tachycardia:
         i. Open suture line.
         ii. Initiate empiric antibiotic therapy targeted against operative site–related suspected pathogens. Examples include:
            (a) Extremity, head, neck, or trunk site: Cefazolin or vancomycin if MRSA suspected
            (b) GI, genitalia, or perineum site: Cephalosporin or levofloxacin plus metronidazole; ertapenem
         iii. Adjust antibiotics according to culture results. Continue for up to 48 hours or until infection is resolved.
         iv. Serial dressing changes
         v. Continue to assess wound for resolution or progression.
   4. Necrotizing fasciitis
      a. Surgical debridement of necrotic tissue serially (i.e., every 24–48 hours) until no further need for debridement
      b. Antibiotic therapy is empiric or definitive.
         i. Empiric antibiotic therapy should be initiated as early as possible.
         ii. Polymicrobial infection should be suspected, with empiric therapy that is active against aerobic and anaerobic gram-positive and gram-negative pathogens
            (a) Vancomycin or linezolid, plus:
            (b) Piperacillin/tazobactam, broad-spectrum carbapenem, or cefepime plus metronidazole
            (c) Consider clindamycin to decrease pathogenic toxin and cytokine production if *S. pyogenes* is suspected. Linezolid has also been shown to decrease toxin production.
         iii. Empiric antibiotic therapy should be de-escalated according to final culture results.
         iv. Definitive antibiotic therapy should be based on final culture results and antibiotic susceptibility. Infection caused by *S. pyogenes* should be treated with an active β-lactam or vancomycin (in severe penicillin allergy) plus clindamycin. Use of intravenous immunoglobulin (IVIG) is controversial. Limited evidence suggests a shorter time to no further need for debridement but no effect on mortality.
         v. Antibiotic therapy should be continued until surgical debridement is no longer necessary and until resolution of infection-related signs and symptoms.
   5. Organ or space infection: See individual chapters, sections, or guidelines for managing organ/space-specific infection (e.g., pancreatitis, intra-abdominal infection; genitourinary tract).

Patient Case

Questions 15 and 16 pertain to the following case.
R.J. is a 57-year-old man admitted to the SICU after a bowel resection following an acute bowel obstruction. On postoperative day 3, R.J. has worsening tachycardia and decreased urinary output together with a maximum temperature of 102.9°F (39.4°C) and a WBC of 18.2 x 10^3 cells/mm^3. R.J. reports worsening pain around his large abdominal surgical wound despite minimal erythema. On opening a few sutures of the wound, pus is expressed on examination, and crepitus is noted in the area surrounding the wound. R.J. is emergently taken to the operating room for exploration and is noted to require significant debridement of necrotic subcutaneous tissue, including the involved fascia. R.J.'s wound is left open with a temporary gauze dressing.

15. Which empiric antimicrobial regimen is most appropriate for R.J.?
    A. Ceftaroline and vancomycin.
    B. Clindamycin, penicillin G, and vancomycin.
    C. Piperacillin/tazobactam, clindamycin, and vancomycin.
    D. Clindamycin and linezolid.

16. Intraoperative tissue culture from the first debridement of R.J.'s wound shows many gram-positive cocci in chains and moderate non–lactose-fermenting gram-negative bacilli. R.J. is to undergo serial intraoperative debridements of necrotic tissue. Given the concern for *S. pyogenes* from the Gram stain, which pharmacotherapeutic intervention would be most appropriate for treatment of R.J.'s necrotizing fasciitis?
    A. Add synergistic gentamicin.
    B. Continue or initiate adjunctive clindamycin for toxin production.
    C. Ensure vancomycin is part of the regimen for concerns of *S. pyogenes* resistance.
    D. Give intravenous immunoglobulin (IVIG).

VIII. STEVENS-JOHNSON SYNDROME AND TOXIC EPIDERMAL NECROLYSIS

A. Epidemiology
    1. Severe cutaneous reactions and related syndromes are unpredictable and rare. The primary causes of these injuries include drugs, dysregulated immune response, and acute infection. Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN also referred to as Lyell's Syndrome), and drug hypersensitivity syndrome, or drug rash with eosinophilia and systemic symptoms (DRESS) are the most common presentations. There is a suggestion of genetic influence on the occurrence of SJS and TEN.
    2. SJS and TEN are the most severe of reactions, representing different points on a similar spectrum of cutaneous injury involving epidermolysis or separation of the epidermis from the dermis.
    3. Similar to thermal injury, the TBSA affected is used to describe the extent of cutaneous injury. According to contemporary reports from U.S. burn centers, the mean TBSA involvement for patients with TEN is greater than 60%.
    4. The incidence of severe cutaneous reactions is difficult to estimate. The highest rates reported approach 20%; however, definitions include mild to moderate reactions. The estimated incidence of SJS and TEN is 1 in 500,000–1,000,000 population. Patients infected with HIV may have a higher incidence.

5. The severity of clinical presentation is associated with the extent of tissue and mucosal injury and necrosis. Most patients have a prodromal fever and malaise preceding cutaneous symptoms. Clinical presentation usually includes fever, SIRS, hypotension from cytokine-mediated vasodilation, and mild to moderate hypovolemia from volume depletion and third spacing. Bleeding may also be present, depending on the extent of mucosal injury.
6. Health care–associated or nosocomial complications, including pneumonia, CAUTIs, CLABSIs, and malnutrition, are common in severely injured patients.
7. SJS and TEN are life-threatening reactions. Average crude mortality associated with SJS/TEN is 25%–55% and can be as high as 90%.

B. Definitions
1. SJS and TEN are part of the same disease process, differing in severity. Common features of SJS and TEN include cutaneous erythema, progressive blistering, epidermolysis, and mucosal erosions.
2. The most widely accepted classification system for SJS and TEN was proposed by Bastuji-Garin et al. This system includes five categories:
   a. Bullous erythema multiforme: Epidermal detachment involving less than 10% TBSA and localized typical targets or raised atypical targets
   b. SJS: Epidermal detachment involving less than 10% TBSA and widespread erythematous or purpuric macules or flat atypical targets
   c. SJS/TEN overlap: Epidermal detachment involving 10%–30% TBSA and widespread purpuric macules or flat atypical targets
   d. TEN with spots: Epidermal detachment involving greater than 30% TBSA and widespread purpuric macules or flat atypical targets
   e. TEN without spots: Large sheets of epidermal detachment involving greater than 10% TBSA without purpuric macules or target lesions

C. Etiology
1. Drugs are the most common cause of SJS and TEN and are implicated in more than 90%–95% of cases. More than 200 medications have been reported as causing SJS and TEN. The most common agents implicated are sulfonamide antibiotics and aromatic anticonvulsants (phenytoin, phenobarbital, and carbamazepine). Other agents/classes include:
   a. Abacavir
   b. Allopurinol
   c. β-Lactam antibiotics
   d. Lamotrigine
   e. Nevirapine
   f. Nonsteroidal anti-inflammatory drugs, particularly the oxicams
   g. Quinolones, particularly ciprofloxacin
   h. Tetracyclines
2. Vaccinations also have been associated with SJS and TEN, including measles, mumps, and rubella (MMR).
3. Exposure to industrial chemicals and fumigants
4. Infection with *Mycoplasma pneumoniae*

D. Diagnosis
   1. Primary clinical signs/symptoms associated with SJS and TEN are fever and malaise, followed by cutaneous blisters and erosive mucosal lesions of the mouth, lips, eyes, and genital area. The distribution of cutaneous lesions is predominantly central, with mucosal involvement of usually at least two sites. Lesions consist of widespread, flat atypical targets or purpuric macules. TBSA is used to differentiate SJS from TEN and to categorize the severity of cutaneous and mucosal involvement. TBSA is calculated by the following:
      a. Arms – 9% each
      b. Head and neck – 9%
      c. Legs – 18% each
      d. Perineum – 1%
      e. Trunk – Anterior 18%; posterior 18%
   2. The diagnosis of SJS and TEN is confirmed by histopathologic analysis of lesional tissue and is corroborated with clinical presentation. Early lesions show scattered necrotic keratinocytes in the epidermis, whereas late-stage lesions reveal confluent full-thickness epidermal necrosis, which leads to formation of subepidermal bullae.
      a. SCORTEN is a severity-of-illness system designed to predict mortality for TEN. It is computed within the first 24 hours of presentation and on day 3 using the sum of seven objective clinical variables (each item present is worth 1 point):
         i. Age older than 40 years
         ii. Heart rate greater than 120 beats/minute
         iii. Presence of cancer or hematologic malignancy
         iv. Epidermal detachment greater than 10% TBSA on day 1
         v. Blood urea nitrogen greater than 28 mg/dL
         vi. Glucose greater than 252 mg/dL
         vii. Serum bicarbonate less than 20 mEq/L
      b. Mortality prediction increases sharply with each additional point, starting at 3% for 0 or 1 point and reaching 90% for 5 or more points. SCORTEN mortality estimates are often used as benchmark rates to assess noncontrolled pharmacotherapy studies.

E. Management and Treatment
   1. Identification, discontinuation, and avoidance of likely or suspected causes are imperative. Causative agents with a long half-life should be identified and strategies to expedite removal considered.
   2. Transfer to ICU, preferably at a certified burn center.
   3. Overt assessment of mucus membranes to prevent extension of injury and related sequelae. This includes the respiratory tract, eyes, and GI tract.
   4. The cornerstones of management of SJS and TEN are:
      a. Resuscitation and supportive care (Recent supportive care guidelines are outlined in J Am Acad Dermatol 2020;82:1553-67)
         i. Goal-directed fluid resuscitation should be initiated immediately to maintain:
            (a) Mean arterial blood pressure above 65 mm Hg
            (b) Central venous pressure 8–12 mm Hg
            (c) Urine output 0.5–1 mL/kg/hour
            (d) Central venous oxygen saturation above 70%
         ii. Support respiratory function with respiratory therapy and continual assessment for intubation, as appropriate
         iii. Avoid using skin to anchor devices and catheters.
         iv. Physical and occupational therapy when appropriate

b. Limit debridement of necrotic epidermis. Best practice recommendations support leaving detached epidermis in place as a biologic dressing along with coverage of affected areas with artificial or biologic dressing. This may be done serially because progression of affected areas may occur. Topical emollients (e.g., petroleum jelly) should be applied to entire epidermis.
c. Management of extracutaneous injuries
    i. Ocular involvement
        (a) Adequate ocular lubrication
        (b) Consideration of topical, preservative-free ophthalmic corticosteroid drops
        (c) Treatment of corneal fluorescein or ulceration
    ii. Oral involvement
        (a) Maintain lip barrier integrity (e.g., white paraffin)
        (b) Consider antiseptic oral rinse (e.g., chlorhexidine)
        (c) Consider topical corticosteroid rinse (e.g., betamethasone)
d. Nutrition support (see related chapter)
e. Avoidance and treatment of infectious complications
    i. Implement infection prevention best practices, including minimizing unnecessary devices and procedures related to health care–associated infection.
    ii. Prophylactic antibiotic therapy is not recommended for SJS and TEN.
    iii. Empiric antibiotic therapy should be carefully chosen and reserved for suspected infection, as evidenced by signs and symptoms of sepsis or site-specific infection. Continuation of antibiotic therapy should be reserved for confirmed infection, and duration should be limited according to the specific infection.

5. Adjuvant therapies
    a. Plasmapheresis – Support is derived from case series; thought to be generally safe and an effective strategy to remove pathogenic, nondialyzable plasma factors, including some drugs, toxins, metabolites, antibodies, immune complexes, and disease-inducing cytokines
    b. Immunomodulating therapy
        i. Corticosteroids – Despite some evidence of benefit, use is controversial and not universally recommended. More recent case reports suggest that high-dose pulse therapy during the first 3 days of presentation decreases disease progression. Associated risks (e.g., infection; hyperglycemia, poor wound healing) may outweigh benefits.
        ii. Cyclosporine – Information from individual case series suggests benefit at a dose of 3 mg/kg/day. There are no formal recommendations for routine use.
        iii. Cyclophosphamide – Early case reports suggested benefit, but cyclophosphamide is not recommended.
        iv. Colony-stimulating factor – May be used in conjunction with cyclosporine in patients with neutropenia and TEN
        v. IVIG
            (a) IVIG for SJS and TEN is controversial.
            (b) In vitro data support that immunoglobulin G (IgG) antibodies against Fas-FasL proteins may decrease keratinocyte apoptosis.
            (c) Many retrospective single-group and cohort studies suggest benefit (usual dosage 1 g/kg/day for 3 days) over SCORTEN estimated mortality rates and similar control groups, respectively.
            (d) Given the rare incidence and logistical difficulty of designing a multicenter prospective study, available prospective studies have been small and single center. Results from these studies, however, have shown no benefit to trends of worse outcome.
            (e) A systematic review and meta-analysis of use in TEN patients found no benefit over standard of care.

(f) The decision to administer IVIG remains clinically supported by pathophysiology-pharmacology interactions and observational data. Centers with expertise to care for patients with TEN should assess the utility of IVIG and develop interdisciplinary guidance for local use.

vi. Consensus guidelines from the United Kingdom recommend immunomodulating therapies be used under the supervision of skin failure specialists in the context of a clinical study or registry. In contrast, consensus guidelines from India recommend (grade B recommendation) that low-dose IVIG be considered in the first 24–48 hours in patients with HIV, children, and pregnant women in the first trimester.

Patient Case

Questions 17 and 18 pertain to the following case.
L.H. is a 23-year-old woman with a recently diagnosed uncomplicated urinary tract infection. Two days after starting sulfamethoxazole/trimethoprim, L.H. presents to the burn ICU from an outside hospital with severe systemic inflammatory response and 30% total body surface area (TBSA) epidermolysis of her upper arms and back. L.H. reportedly presented to the outside hospital 6 hours earlier with only 10% TBSA involvement. New lesions since admission to the burn ICU include erythema of her thighs and oral cavity with TBSA now at 40%, suggestive of toxic epidermal necrolysis (TEN). L.H.'s SCORTEN score is 3.

17. Which would be the best initial pharmacotherapeutic intervention?
    A. Crystalloid resuscitation.
    B. Cyclophosphamide.
    C. Empiric vancomycin for wound prophylaxis.
    D. High-dose systemic corticosteroids.

18. L.H. continues to worsen, with a TBSA involvement now at 50%, with worsening oral cavity involvement and progressing acute kidney injury. You and your team consider IVIG as the next pharmacotherapeutic intervention. Which best describes the evidence-based role of IVIG in managing TEN?
    A. IVIG should be administered before the patient's condition progresses to TEN.
    B. IVIG should be reserved for specialty centers with interdisciplinary consensus protocols to guide use.
    C. Multicenter pivotal trials show that IVIG is most efficacious for patients with ocular involvement.
    D. Meta-analyses of the available evidence support that IVIG is standard of care in managing TEN.

IX. INFLUENZA

A. Epidemiology
    1. Influenza is a seasonal viral illness affecting all ages and associated with significant morbidity and mortality. Seasonal patterns in the United States vary from year to year. For example, the 2008–2009 influenza season started in November 2008 and waned by April 2009, but then, because of influenza A H1N1 (2009 H1N1), it resurged in May 2009, peaking in mid-October 2009 and persisting to April 2010. The 2022–2023 season started to significantly increase in October 2022 before peaking in December 2022.

2. According to the CDC, it is estimated that 400,000–900,000 patients a year in the United States are hospitalized with influenza-related illness. During the previous 2020–2021 season, the reported overall rate (cumulative rate per 100,000 population) of hospitalization for laboratory-confirmed influenza was 0.8, notably lower than previous seasons (67 for 2019–2020 season). However, the rate increased during the 2021–2022 season to 17.3, which is still notably lower compared with rates prior to the SARS-CoV-2 pandemic. Age-related rates for adult hospitalizations were 50.4 for those 65 years of age and older, 16.0 for 50–64 years of age, and 9.1 for 18–49 years of age. The rate of ICU admission among those hospitalized was 14.3%, compared to the the five seasons from 2015–2020 (range 15%–19%).
3. Risk factors for severe influenza requiring hospitalization include chronic respiratory or metabolic illness, immunosuppression (disease or pharmacotherapy), pregnancy, and age older than 65 years. In 2009, re-emergence of the high-virulence strain influenza A H1N1 (pandemic 2009 H1N1) caused significant morbidity and mortality in young adults and other groups considered at a lower risk of severe influenza. The pandemic 2009 H1N1 strain continues to be tracked and varies significantly from year-to-year. In 2022–2023, the 2009 H1N1 was responsible for approximately 28.3% of reported influenza cases in the United States.
4. The most common complications of severe influenza include hypoxemic respiratory failure, bacterial pneumonia, and ARDS. Concomitant sepsis, septic shock, multiple organ failure, encephalopathy, and rhabdomyolysis also are associated with severe influenza.
5. Among all causes of deaths in the United States, pneumonia and influenza have surpassed the epidemic threshold since 2009, with rates close to 10% during the peak of influenza seasons. Severe influenza caused by the 2009 H1N1 strain has been associated with crude mortality rates of 15%–53%.

B. Etiology
1. Influenza is caused by the RNA viruses influenza A, B, or C, which usually spread through droplet transmission. Influenza A and B viruses are the predominant causes of clinically significant illness. Influenza virus subtypes are described by surface proteins hemagglutinin (H) and neuraminidase (N).
2. According to the CDC, influenza A has been the most prevalent cause of influenza since 2009, with 2009 H1N1 and H3 being the most prevalent influenza A subtypes. Influenza B prevalence has varied during the same period, but it remains an important cause of illness. Overall for 2022–2023, influenza A was isolated in 95.4% of cases. The most common specified influenza strain isolated was influenza A H3N2 at 67.1%.
3. Influenza viruses can cause a broad range of respiratory tract infections, ranging from mild to moderate upper respiratory tract infections to severe pneumonia. Influenza infection has been associated with acute viremia.

C. Prevention
1. Annual vaccination remains the primary tool for influenza prevention.
2. In-hospital and ICU outbreaks of influenza can contribute to viral transmission and associated sequelae.
3. Institutions should begin implementing influenza screening and infection control measures when influenza viruses are confirmed to be in the local community.
4. The CDC recommends the implementation of droplet precautions for hospitalized patients with suspected or confirmed influenza; further, these precautions are recommended for 7 days after illness onset or until 24 hours after the resolution of fever and respiratory symptoms, whichever is longer.

D. Diagnosis
1. Patients with influenza may present with fever, myalgias, headache, malaise, dry cough, pharyngitis, and rhinorrhea. Fever and myalgias generally last 3–5 days, whereas malaise and respiratory symptoms may last 2 weeks or more. Patients with severe influenza may present with hypoxemic respiratory failure and sepsis.

2. Clinical signs and symptoms of influenza are nonspecific. To confirm the diagnosis of influenza, sampling of the upper respiratory tract using nasal washing or nasopharyngeal swab or lower respiratory tract within 5 days of illness is preferred. Diagnostic tests obtained beyond 5 days may have false-negative results.
3. Diagnostic tests available for influenza include viral culture, serology, rapid antigen testing, polymerase chain reaction (PCR), and immunofluorescence assays.
    a. Viral culture and reverse transcription polymerase chain reaction (RT-PCR) are the most sensitive and are the only tests able to identify individual influenza subtypes. These tests can be performed using nasopharyngeal swab or respiratory tract culture (e.g., sputum, BAL).
    b. Rapid diagnostic tests of nasopharyngeal swab or nasal wash have high specificity (90%–95%) (i.e., low false-positive rate and rapid turnaround), promoting these tests as first line for general diagnosis. Depending on the rapid diagnostic assay used, differentiation between influenza A and influenza B is possible, although further subtype identification is not available with these tests. Because of poor sensitivity, negative rapid diagnostic tests should be followed up with viral culture or RT-PCR.
4. Case definitions for influenza surveillance are suspected, probable, or confirmed based on patient presentation and laboratory assessment.
    a. Suspected – Mild to severe influenzalike illness within reasonable seasonal threshold.
    b. Probable – Mild to severe influenzalike illness within reasonable seasonal threshold or after recent contact with person with probable or confirmed influenza infection and not otherwise explained.
    c. Confirmed – Mild to severe influenzalike illness with confirmatory laboratory tests indicating presence of influenza A or B.

E. Management and Treatment
1. Management of critically ill patients with influenza includes treatment of primary influenza infection, secondary bacterial infection(s), and related noninfectious complications (e.g., respiratory failure, ARDS, prolonged mechanical ventilation, multiple organ failure). Management of common bacterial infections and noninfectious complications is described in other chapters.
    a. Diagnostic tests for influenza should be obtained and empiric antiviral treatment of influenza initiated during seasonal outbreaks in critically ill patients presenting with acute febrile and respiratory illness consistent with influenza. Severely ill patients with influenza may present with hypothermia similar to other populations with sepsis. Accompanying severe hypoxemic respiratory failure should heighten the concern for influenza.
    b. Antiviral treatment (Table 4) should be initiated within 48 hours from onset of symptoms. Hospitalized patients receiving antiviral therapy after 48 hours from symptom onset may benefit. The CDC recommends that treatment be initiated even 5 days after symptoms in critically ill patients.
    c. Neuraminidase inhibitors continue to be the mainstay of antiviral therapy for recent influenza A and B strains. These agents inhibit viral neuraminidase, decreasing the release of post-replication viral particles (virions) from infected cells, limiting spread to additional tissues. Neuraminidase inhibition may also suppress the initiation of infection after acute inoculation.
        i. Oseltamivir phosphate is an oral (capsule, powder for suspension) ethyl ester prodrug converted through hepatic ester hydrolysis to the active form oseltamivir carboxylate. Oseltamivir can be administered to critically ill patients by orogastric or nasogastric tube. An intravenous formulation of oseltamivir has been tested in early clinical trials. Oseltamivir is the drug of choice for hospitalized patients with severe influenza.

ii. Zanamivir is available on the market as an orally inhaled drug delivered by a Diskhaler device. Intravenous zanamivir was once available for patients with severe influenza through clinical trial participation or emergency investigational new drug through the manufacturer. However, this program has been discontinued for countries outside the European Union (internal communication).

iii. Peramivir is an intravenous neuraminidase inhibitor with activity against influenza A and B. Peramivir was approved in December 2014 for use in patients 18 years or older with acute uncomplicated influenza. Peramivir is administered as a single 15-minute intravenous infusion. Data are limited for peramivir in critically ill patients.

d. Adamantanes inhibit the replication of influenza A viruses by preventing viral assembly. Adamantanes also interfere with the function of the transmembrane domain of the viral M2 protein, preventing release of viral particles into host cells.

e. Baloxavir is an endonuclease inhibitor approved by the FDA in 2018 for the management of uncomplicated influenza in individuals older than 12 years. Baloxavir inhibits endonuclease of influenza virus A and B strains, limiting the activity of the polymerase acidic protein responsible for viral gene transcription and virus replication. Data are limited on its safety and efficacy in patients with severe influenza.

f. Susceptibility of influenza to available antiviral agents varies from year to year and between strains and subtypes. Local surveillance of susceptibility patterns at the beginning and throughout a respective season is imperative to provide appropriate therapy.

i. In the past three seasons, 100% of tested influenza A (2009 H1N1 and H3) and B viruses were susceptible to oseltamivir, peramivir, and zanamivir.

ii. High levels of resistance to the adamantanes (amantadine and rimantadine) have persisted among circulating influenza A viruses. The adamantanes are not effective against influenza B viruses; therefore, they are not viable options for the management of contemporary influenza.

Table 4. Influenza-Specific Pharmacotherapy

Agent	Drug Class; Influenza Activity	Bioavailability	Half-life (hr)	Treatment Dosage	Adverse Effects	Comments
Oseltamivir phosphate	Neuraminidase inhibitor; A and B	Orally or OG/NG tube: > 75%	6–10	75 mg twice a day for 5 days[a]; up to 150 mg twice a day has been used in patients with severe influenza but is not associated with improved outcomes and thus is not currently recommended by the CDC	Nausea, vomiting, delirium	Dosage adjustment for renal impairment may be necessary; patients requiring continuous renal replacement therapy should receive normal dosing

Table 4. Influenza-Specific Pharmacotherapy *(continued)*

Agent	Influenza Activity	Bioavailability	Half-life (hr)	Treatment Dosage	Adverse Effects	Comments
Peramivir	Neuraminidase inhibitor; A and B; limited clinical trial experience for Influenza B strains	IV: 100%	20	600 mg once for 15 minutes	Diarrhea, hypersensitivity reactions, delirium	Dosage adjustment recommended in patients with creatinine clearance below 50 mL/min; should be administered after dialysis in patients receiving hemodialysis
Zanamivir	Neuraminidase inhibitor; A and B	Inhaled: Up to 17%	2.5–5	10 mg twice a day for 5 days[a]	Allergic reactions, diarrhea, nausea, headache, dizziness	Not recommended in patients with chronic lung disease or severe influenza; not available for delivery through mechanical ventilator circuit because of bronchospasm
Baloxavir marboxil	Polymerase acidic endonuclease inhibitor; A and B	Tablet, oral: Prodrug converted almost completely to active baloxavir; bioavailability not formally established	79% (53%–96%); primary hepatic metabolism (80% of dose)	Weight 40–79.9 kg: 40 mg orally once; weight ≥ 80 kg: 80 mg orally once; no information available for feeding tube administration	Generally minimal (< 5% of patients): Diarrhea, bronchitis, nausea, sinusitis, headache	93% protein binding; limited data for hospitalized patients with severe influenza
Amantadine	Adamantane; A only	70%–100%	15–17	100 mg twice a day	Nausea, dizziness, insomnia, lower seizure threshold, anticholinergic effects	High resistance to current influenza A strains

Table 4. Influenza-Specific Pharmacotherapy *(continued)*

Agent	Influenza Activity	Bioavailability	Half-life (hr)	Treatment Dosage	Adverse Effects	Comments
Rimantadine	Adamantane; A only	> 90%	24–35	100 mg twice a day	Dizziness, nausea, vomiting; anticholinergic effects	High resistance to current influenza A strains; limited market availability

[a]Longer durations of up to 14 days may be needed for severely ill patients.

IV = intravenous(ly); NG = nasogastric; OG = orogastric.

2. Response to antiviral therapy should be assessed throughout treatment using clinical signs and symptoms of infection. Development of oseltamivir resistance during therapy has been reported, but this is rare. If suggested through local surveillance, therapy should be switched to zanamivir by the appropriate administration route. Improvement in infectious and noninfectious complications may be delayed, despite resolution of primary influenza infection. ARDS, in particular, may persist for days beyond primary influenza.

Patient Cases

19. T.Y., a 32-year-old woman who has had flulike symptoms for the past 72 hours, presents to the ED from home with severe fatigue, shortness of breath, and rigors. T.Y. has a heart rate of 120 beats/minute, mean arterial pressure 70 mm Hg, respiratory rate 24 breaths/minute, temperature 102.7°F (39.3°C), and arterial oxygen saturation (SaO_2) of 85% on room air. Because it is in the middle of influenza season (high prevalence of influenza A and B), a nasal swab is done and sent for rapid diagnostic testing for suspected influenza infection. Shortly thereafter, T.Y. is intubated for severe respiratory failure and admitted to the MICU. In addition to antibiotic therapy for CAP, which would best be considered next?
    A. None; the patient is outside the time window to effectively treat influenza.
    B. Await rapid diagnostic test results before initiating influenza-specific therapy.
    C. Give amantadine.
    D. Give oseltamivir.

20. G.L. is a 25-year-old woman admitted to the medical ICU in late February this year with severe, acute hypoxemic respiratory failure requiring mechanical ventilation. G.L. is hypotensive, requiring norepinephrine after fluid resuscitation, and has acute kidney injury. She is awaiting a renal consult for continuous renal replacement therapy. A rapid diagnostic test of a nasal washing suggests influenza B, consistent with this year's local influenza epidemiology. A nasogastric feeding tube is placed when she is admitted to the ICU. Which would be most appropriate to treat G.L.'s severe influenza?
    A. Enteral amantadine.
    B. Enteral oseltamivir.
    C. Inhaled zanamivir.
    D. Intravenous peramivir.

X. SEVERE ACUTE RESPIRATORY SYNDROME CORONAVIRUS 2

A. Epidemiology
   1. In late December 2019, a novel coronavirus strain causing pneumonia in a human was first identified in Wuhan, China. This virus, first labeled 2019-nCoV, is now called novel severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2).
   2. As of March 10, 2023, there have been more than 676 million cases of SARS-CoV-2 worldwide, with over 6.8 million associated deaths. More than 103 million cases and over 1 million deaths have been attributed to SARS-CoV-2 in the United States (https://coronavirus.jhu.edu/map.html).
   3. Illness related to SARS-CoV-2 is called coronavirus disease 2019 (COVID-19). COVID-19 was categorized as a worldwide pandemic on March 11, 2020.
   4. COVID-19 is characterized by a broad spectrum of illnesses, from mild upper respiratory symptoms to life-threatening respiratory failure, sepsis, and/or multiorgan failure.
   5. Before vaccination and other SARS-CoV-2-directed therapeutics, about 80% of patients with COVID-19 present with mild illness, and 15%–20% present with more severe illness requiring hospitalization. Between 2% and 10% of patients with COVID-19 require ICU admission, accounting for 30%–40% of patients hospitalized with COVID-19.
   6. Risk factors for COVID-19–related hospitalization span several comorbidities, including hypertension, diabetes, cardiovascular disease, COPD, obesity, and chronic kidney disease. Patients with comorbidities are at increased risk of hospitalization, ICU admission, and mortality.
      a. The most common complication of COVID-19 in ICU patients is hypoxemic respiratory failure from viral pneumonia, usually bilateral. Cytokine release syndrome or "cytokine storm" with multiorgan failure may be the primary presentation in some patients.
      b. Between 25% and 90% of patients admitted to the ICU require MV.
      c. Additional acute complications: Septic shock, liver dysfunction, bleeding and coagulation dysfunction, cytokine release syndrome, lymphopenia, and acute cerebrovascular disease
      d. Late complications: Myocarditis, cardiomyopathy, and ventricular arrhythmias
   7. In-hospital crude mortality for COVID-19 is generally 15%–60%. Rates have declined in areas with higher levels of immunity.
      a. Mortality approaches 40% in ICU patients with COVID-19.
      b. Age-related in-hospital mortality ranges from less than 5% for individuals younger than 40 to greater than 60% for those older than 80.

B. Etiology
   1. SARS-CoV-2 is a coronavirus; coronaviruses are relatively large, enveloped, single-stranded RNA viruses. Clinically, coronaviruses are usually associated with the common cold and mild to moderate gastrointestinal illness. Common seasonal strains include coronaviruses 229E, HKU1, NL63, and OC43.
   2. SARS-CoV-2, SARS-CoV (SARS), and Middle East respiratory syndrome (MERS) are the most recent coronaviruses to cause severe respiratory infection.
   3. SARS-CoV-2 predominantly targets nasal and bronchial epithelial cells and lower respiratory tract epithelial pneumocytes.

4. SARS-CoV-2 pathogenicity is mediated through binding of the viral structural spike protein (S protein) to membrane angiotensin-converting enzyme 2 (ACE2) receptors, predominant in alveolar type II cells. Subsequently, serine protease type 2 transmembrane serine protease (TMPRSS2) cleaves the ACE2 receptor–S protein complex, leading to activation of the viral S protein and enhanced cellular uptake. Once intracellular, SARS-CoV-2 RNA is incorporated into the host nucleus, and viral replication ensues. Activation of the ACE2 receptor may also implicate the contact pathway and complement system, perhaps through an ACE2-mediated increase in bradykinin, in the pathophysiology of COVID-19. This has been hypothesized to be the primary mechanism of cytokine release syndrome in patients with COVID-19.
5. Mutations in SARS-CoV-2 continue to be reported.
    a. One of the first reported variants was D614G, which has a modified S protein that may increase cellular viral uptake. Many areas worldwide are concurrently experiencing multiple SARS-CoV-2 variants.
    b. To date, the Delta variant (B.1.617.2) has demonstrated increased transmissibility (about 2-fold) and pathogenicity compared with the wild-type SARS-CoV-2 strain.

C. Prevention
1. SARS-CoV-2 is primarily transmitted from human to human through respiratory secretions, most likely through droplets. Transmission is also possible through direct surface contact, though this is less likely.
2. Physical distancing, handwashing, and airborne pathogen precautions (i.e., mask wearing) together with contact precautions in medical environments are recommended to decrease transmission.
3. Clinicians caring for patients undergoing aerosol-generating procedures (e.g., endotracheal intubation; bronchoscopy) should don respirator masks together with eye protection and full-body gowning. Negative-pressure patient rooms are preferred during aerosol-generating procedures.
4. The incubation period for SARS-CoV-2 is estimated to be up to 14 days from exposure. Information on onset and duration of viral shedding and infectious potential is still emerging and suggests variable time intervals of 2–18 days from symptom onset. Individuals with symptoms likely have a higher potential for transmitting the virus.
5. Several vaccines against SARS-CoV-2 are available and should be considered per CDC recommended schedules.

D. Diagnosis
1. Primary COVID-19 symptoms in hospitalized patients include fever (80%), dry cough (75%), shortness of breath (70%), fatigue (40%), myalgias (35%), and gastrointestinal symptoms (30%). The most common reason for hospitalization is shortness of breath or hypoxemia.
2. Laboratory abnormalities in hospitalized patients with COVID-19 include:
    a. Lymphopenia (related to T-cell lysis from direct infection by SARS-CoV-2)
    b. Elevated inflammatory markers (e.g., C-reactive protein, erythrocyte sedimentation rate, ferritin, interleukin [IL]-6, tumor necrosis factor alpha). May resemble cytokine release syndrome
    c. Coagulation abnormalities (e.g., prolonged prothrombin time, thrombocytopenia, elevated D-dimer, hypofibrinogenemia)
3. Radiographic abnormalities can vary, but patients with severe disease commonly present with bilateral, lower lobe–predominant infiltrates on chest radiography. Chest CT can reveal bilateral, ground-glass opacities with or without consolidation, usually depending on the time course of presentation.
4. The CDC recommends that nasopharynx samples be used to detect SARS-CoV-2 in patients presenting with COVID-19 symptoms. Lower respiratory tract samples (e.g., endotracheal aspirate, BAL) have a higher diagnostic yield, but concerns with aerosolization and lack of tolerability in those with severe hypoxemia may limit their usefulness.
5. COVID-19 diagnosis is confirmed using RT-PCR.

E. Management and Treatment
   1. Management priorities for critically ill patients with COVID-19 include supportive care of hypoxemic respiratory failure and septic shock; immunomodulation of the inflammatory coagulation response, including consideration for anticoagulation; and primary antiviral therapy.
   2. New information is emerging on effective management strategies and therapies. Useful resources include the National Institutes of Health COVID-19 guidelines (www.covid19treatmentguidelines.nih.gov) and the Surviving Sepsis Campaign (Crit Care Med 2020;48:e440-e469).
      a. Hypoxemic respiratory failure
         i. Supplemental oxygen to maintain SaO_2 above 92%–96%; high-flow nasal cannula preferred to noninvasive positive pressure ventilation
         ii. For patients who require MV: ARDSNet lung-protective strategy
            (a) Low tidal volumes
            (b) Appropriate plateau pressures
            (c) Higher PEEP (positive end-expiratory pressure)
            (d) Intermittent or continuous (in those for whom intermittent dosing fails) neuromuscular blockade
            (e) Proning
         iii. Extracorporeal membrane oxygenation (ECMO): Venovenous ECMO in patients with refractory hypoxemia despite optimizing ventilation, rescue therapies, and proning
      b. Septic shock
         i. Goal blood pressure: MAP of 60–65 mmHg
         ii. Fluid resuscitation: Conservative management strategy using balanced crystalloid solutions
         iii. Vasopressors: Norepinephrine as first line; vasopressin or epinephrine as second line
         iv. Inotropes: Dobutamine if cardiac dysfunction or persistent hypoperfusion despite fluid resuscitation and norepinephrine
         v. Corticosteroids: See Immunomodulation section in the text that follows.
         vi. Empiric antibiotic therapy should be reserved for patients with confirmed of high suspicion of concomitant bacterial infection.
      c. Anticoagulation without contraindication or other compelling indication for therapeutic anticoagulation
         i. Hospitalized patients not requiring oxygen should receive prophylactic heparin
         ii. In non-ICU and non-pregnant hospitalized patients requiring conventional oxygen therapy AND with a d-Dimer above the upper limit of normal should receive therapeutic heparin for 14 days or until they are transferred to the ICU or discharged from the hospital, whichever comes first. Pregnant patients should receive prophylactic heparin as appropriate.
         iii. Hospitalized patients requiring high-flow oxygen, noninvasive or invasive mechanical ventilation, or ECMO should receive prophylactic heparin.
      d. Immunomodulation
         i. Corticosteroids
            (a) Dexamethasone (or equivalent-dose prednisone, methylprednisolone, or hydrocortisone) 6 mg per day orally or intravenously for up to 10 days or until hospital discharge is recommended for COVID-19 treatment in hospitalized patients requiring increased amounts of supplemental oxygen or MV or ECMO.
               (1) The largest corticosteroid study to date is the RECOVERY trial, a randomized, open-label trial comparing dexamethasone 6 mg orally or intravenously once daily for up to 10 days with usual care alone in 6425 patients (dexamethasone 2104; usual care 4321) hospitalized with COVID-19. At baseline, 3883 patients required supplemental oxygen, and 1007 required MV.

(A) Dexamethasone decreased 28-day mortality (primary outcome) (dexamethasone 22.9% vs. usual care 25.7%; age-adjusted rate ratio 0.83; 95% CI, 0.75–0.93; p<0.001).
(B) Results were similar among patients who required MV (29.3% vs. 41.4%; rate ratio 0.64; 95% CI, 0.51–0.81) and those receiving supplemental oxygen (23.3% vs. 26.2%; rate ratio 0.82; 95% CI, 0.72–0.94). There was no difference among patients not requiring respiratory support.

(b) A meta-analysis of seven controlled trials encompassing 1703 critically ill patients (requiring supplemental oxygen or MV) with COVID-19, including 1007 patients (15.7% of end enrollment) enrolled early in RECOVERY, showed that corticosteroids were associated with lower 28-day all-cause mortality compared with placebo or usual care (OR 0.66 [95% CI, 0.53–0.82]; p<0.001).

(c) The prospective, randomized, controlled COVID STEROID 2 trial evaluated dexamethasone 6 mg or 12 mg daily for up to 10 days in 982 hospitalized patients.
(1) Median number of days alive without life support at 28 days was 20.5 days (interquartile range [IQR] 4.0–28.0) vs. 22.0 days (IQR 6.0–28.0) (adjusted mean difference 1.3 days (95% CI, 0–2.6; p=0.07).
(2) No difference was found in 28-day or 90-day mortality.

ii. Other immunomodulators

(a) Toclizumab: Anti-IL-6 receptor monoclonal antibody (alternative class agent: sarilumab)
(1) The COVACTA trial (N Engl J Med 2021;384:1503-16) was a randomized, placebo-controlled trial in 438 hospitalized adults with severe COVID-19 pneumonia.
(A) Patients were randomized 2:1 to either a single intravenous infusion of tocilizumab (8 mg/kg actual body weight) or placebo; 25% of patients received a second dose of tocilizumab.
(B) At enrollment, 37% of patients required MV and 22.4% received concomitant corticosteroids.
(C) No between-group differences were observed in clinical progression or 28-day mortality (tocilizumab, 19.7% vs. placebo, 19.4%; weighted difference 0.3%; 95% CI, –7.6 to 8.2).
(2) The REMAP-CAP trial (N Engl J Med 2021;384:1491-502) randomized 803 patients within 24 hours of ICU admission (median 1.2 days from hospitalization) to open-label tocilizumab 8 mg/kg actual body weight (max of 800 mg) (353 patients), sarilumab 400 mg (48 patients), or standard of care (402 patients).
(A) Most patients (552 [68.7%]) were free of MV and/or ECMO at enrollment; 76% of patients received concomitant corticosteroids.
(B) Tocilizumab was associated with reduced in-hospital mortality (28% vs. 36%; OR 1.64; 95% CI, 1.25–2.14) and higher median days free of respiratory and cardio-vascular organ failure (10 days vs. 0 days; OR 1.64; 95% CI, 1.25–2.14).
(C) Tocilizumab was also associated with decreased progression to MV, ECMO, or death (OR 0.59; 95% CI, 0.41–0.86).
(3) The RECOVERY trial (Lancet 2021;397:1637-45) was a randomized, controlled, open-label study in 4116 hospitalized patients with COVID-19 demonstrating acute hypoxia and evidence of systemic inflammation (i.e., C-reactive protein ≥75 mg/L).
(A) Patients were randomized 1:1 ratio to usual care vs. usual care plus tocilizumab 400 mg–800 mg intravenously once; 29% of tocilizumab patients received a second dose at provider discretion.

(B) At enrollment, 82% of patients received concomitant corticosteroids and 13.8% required MV.
(C) Tocilizumab was associated with a decreased 28-day mortality overall (tocilizumab, 31% vs. usual care, 35%; rate ratio 0.85; 95% CI 0.76–0.94) and across prespeci-fied subgroups based on illness severity.
(D) Patients not requiring MV at enrollment who received tocilizumab were less likely to progress to MV or death (35% vs 42%; risk ratio 0.84; 95% CI 0.77–0.92).

(4) Tocilizumab is authorized for use under an FDA Emergency Use Authorization (EUA) for treatment of COVID-19 in hospitalized patients from 2 to less than 18 years of age who are receiving systemic corticosteroids and require supplemental oxygen, non-invasive or invasive mechanical ventilation, or ECMO. On December 21, 2022, the FDA approved tocilizumba for the treatment of COVID-19 in hospitalized adults who are receiving systemic corticosteroids and require supplemental oxygen, non-invasive or invasive mechanical ventilation, or ECMO.
(5) Possible adverse effects include serious infections, fungal pneumonia, gastrointestinal perforation, hepatotoxicity, and cytopenias.

(b) Baricitinib: Janus kinase inhibitor (alternative class agent: tofacitinib)
(1) The ACTT-2 study (N Engl J Med 2021;384:795-807) was a double-blind, randomized, placebo-controlled trial evaluating baricitinib 4 mg/day orally or via nasogastric tube plus remdesivir (200 mg on day one, then 100 mg daily for 5-10 days) versus placebo in 1033 hospitalized patients with COVID-19.
(A) At enrollment, 32% of patients had severe disease, with 10.7% of patients requiring MV or ECMO. Patients receiving corticosteroids for COVID-19 were excluded.
(B) Overall, patients receiving baricitinib plus remdesivir had a shorter time to recovery (7 days; 95% CI 6–8 vs. 8 days; 95% CI 7–9; rate ratio 1.16; 95% CI, 1.01–1.32), which was most pronounced in patients receiving high-flow oxygen or noninvasive ventilation at enrollment (10 days vs. 18 days; rate ratio 1.51; 95% CI, 1.10–2.08).
(C) No difference was noted in 28-day mortality (5.1% vs. 7.8%; hazard ratio 0.65; 95% CI 0.39–1.09).

(2) The COV-BARRIER trial (Lancet Respir Med 2021; www.thelancet.com/journals/lanres/article/PIIS2213-2600(21)00331-3/fulltext) was a multinational, placebo-controlled, randomized trial in 1525 hospitalized patients with COVID-19 pneumonia and elevated inflammatory markers. Patients receiving MV were excluded.
(A) Patients were randomized 1:1 to receive baricitinib 4 mg/day (renally-adjusted as necessary) orally or placebo in addition to usual care; 92% of patients received concomitant dexamethasone.
(B) No between-group difference were noted in disease progression.
(C) Patients receiving baricitinib overall had lower 28-day all-cause mortality (8% vs. 13%; hazard ratio 0.57; 95% CI 0.41–0.78).
(D) The treatment effect for mortality was most notable among the subgroup of 370 pa-tients receiving high-flow oxygen or noninvasive ventilation at baseline (17.5% vs. 29.4%; hazard ratio 0.52; 95% CI, 0.33–0.80).

(3) The COV-BARRIER Addendum was an exploratory randomized, placebo-controlled trial evaluating baricitinib in 101 patients with confirmed COVID-19 and 1 or more elevated inflammatory markers (CRP, D-dimer, LDH, or ferritin) and MV or ECMO (Lancet Respir Med 2022;10:327-36).

(A) Patients were randomized 1:1 to receive baricitinib 4 mg/day (renally-adjusted as necessary) orally or placebo in addition to usual care; 86% of patients received concomitant dexamethasone.
(B) 28-day mortality: baricitinib 39% versus placebo 58% (hazard ratio 0.54; 95% CI, 0.31–0.96; p=0.030). Baricitinib relative reduction in mortality similar as seen in less severely ill population from COV-BARRIER parent trial.
(C) Number of ventilator-free days and duration of hospitalization: no significant difference between arms

(4) Baricitinib use is supported by an FDA EUA for treatment of COVID-19 in hospitalized patients from 2 to less than 18 years of age requiring supplemental oxygen, non-invasive or invasive MV, or ECMO. Baricitinib was approved on May 10, 2022, for the treatment of COVID-19 in hospitalized adults requiring supplemental oxygen, non-invasive or invasive mechanical ventilation, or ECMO.
(5) Baricitinib can be administered orally or via oral dispersion or gastrostomy or nasogastric tube.
(6) Possible adverse effects include elevations of transaminases, fungal pneumonia, thrombosis, and cytopenias.

(c) Although preclinical models and early clinical reports have been enthusiastic, data are insufficient to support routine use of other immunomodulators in patients with COVID-19. Ongoing clinical trials are anticipated. Additional drug classes/agents reported include:
(1) IL-1 inhibitors (e.g., anakinra)
(2) Interferon alfa or beta
(3) Anti–IL-6 monoclonal antibody siltuximab
(4) Bruton tyrosine kinase inhibitors (e.g., acalabrutinib, ibrutinib, zanubrutinib)

e. Antiviral therapy
i. Remdesivir
(a) Nucleotide prodrug of adenosine analog that binds to viral RNA polymerase, terminating RNA transcription and inhibiting viral replication. Initially developed to treat MERS
(b) FDA-approved as Veklury for adult and pediatric (12 years or older and weighing at least 40 kg) patients hospitalized with COVID-19
(1) Dosage: 200 mg intravenously once, then 100 mg intravenously daily for 5 days; 10 days' duration can be considered in patients with no clinical improvement after 5 days
(2) Precautions include patients with baseline liver dysfunction, particularly those with elevated alanine aminotransferase.
(c) Recommended in hospitalized patients who require supplemental oxygen with or without corticosteroids or those patients at high risk for decompensation or immunocompromised.
(1) Should only be used in combination with corticosteroids in patients requiring high-flow oxygen; not recommended in patients requiring MV or ECMO because benefit is uncertain in subgroups of more severely ill patients.
(2) If supplies are limited, prioritize use for patients not requiring high-flow oxygen; not recommended in patients requiring MV or ECMO.
(d) To date, the largest published, placebo-controlled trial is the Adaptive COVID-19 Treatment Trial (ACTT-1), which compared remdesivir with placebo in 1063 patients hospitalized with COVID-19 and receiving standard of care.

(1) Primary outcome was the time to recovery within 28 days, defined as the first day of clinical improvement to categories 1–3 on an ordinal scale ranging from 1, not hospitalized, return to usual care; to 3, hospitalized, not requiring supplemental oxygen; to 8, death. All patients had scores of 4–7 at enrollment. At baseline, 197 patients (18.5%) required high-flow supplemental oxygen (score of 6), and 292 (25.6%) required MV or ECMO (score of 7).
(2) Patients receiving remdesivir had a shorter time to recovery (median 11 days vs. 15 days; rate ratio 1.32; 95% CI, 1.12–1.55; p<0.001). Among patients with baseline scores of 6 and 7, rate ratios for time to recovery were 1.20 (95% CI, 0.79–1.81) and 0.95 (95% CI, 0.64–1.42), respectively.
(3) Rate ratio for 14-day mortality among all patients was 0.70 (95% CI, 0.47–1.04), with the greatest treatment effect in the largest patient subgroup (n=421) with a baseline score of 5 (rate ratio 0.22 [95% CI, 0.08–0.58]). Corresponding rate ratios among patients with baseline scores of 6 and 7 were 1.12 (95% CI, 0.53–2.38) and 1.06 (95% CI, 0.59–1.92), respectively.
(4) Adverse effects were generally similar between groups.

(e) The prospective, randomized, open-label WHO SOLIDARITY Trial compared remdesivir to standard of care in 8275 hospitalized patients with COVID-19 requiring supplemental or high-flow oxygen (71%) or mechanical ventilation (9%) (Lancet 2022;399:1941-53).
(1) Patients were randomized 1:1 to remdesivir standard dosing for 10 days or standard of care; 68% of patients received corticosteroids.
(2) In-hospital mortality: remdesivir 14.5% versus standard of care 15.6 (rate ratio 0.91; 95% CI, 0.82–1.02; p=0.12)
(3) Patients receiving oxygen: 14.6% versus 16.3% (rate ratio 0.87; 95% CI, 0.76–0.99; p=0.03)
(4) Progression to mechanical ventilation: 14.1% versus 15.7% (rate ratio 0.88; 95% CI, 0.77–1.00; p=0.04)

ii. Hydroxychloroquine or chloroquine
(a) Antimalarial agents that, in vitro, increase the endosomal pH, inhibiting fusion of SARS-CoV-2 to the host cell membranes
(b) Not recommended as definitive or prophylactic therapy because larger controlled trials have failed to show a difference in clinical recovery or mortality in hospitalized patients with COVID-19
(c) Combination with azithromycin reserved for clinical trial

iii. Miscellaneous agents
(a) Ivermectin: Not recommended; evidence does not support use for patients with COVID-19
(b) Lopinavir/ritonavir and other HIV protease inhibitors: Not recommended outside clinical trial

Table 5. National Institutes of Health Guideline Recommendations for Pharmacologic Management of Patients Hospitalized with Coronavirus Disease 2019

Disease Severity	Remdesivir	Dexamethasone[a]	Baricitinib	Tocilizumab[b]	Anticoagulation
No supplemental oxygen requirement	Consider patients at high risk for worsening	Against use	N/A	N/A	For patients without an indication for therapeutic anticoagulation: Prophylactic dose of heparin, unless contraindicated (AI)
Supplemental oxygen required	Recommended in patients with *minimal oxygen requirement*	Recommended in combination with remdesivir in most patients requiring oxygen; may use alone if remdesivir is unavailable	Patients who have rapidly increasing oxygen needs and systemic inflammation, promptly add either agent (rating recommendation/ evidence level: baricitinib BIIa; tocilizumab BIIa) to dexamethasone ± remdesivir in most patients		For nonpregnant patients with D-dimer levels above the ULN who do not have an increased bleeding risk: Therapeutic dose of heparin (CIIa) For other patients: Prophylactic dose of heparin, unless contraindicated (AI)
High-flow oxygen device or noninvasive ventilation requirement	Recommended in combination with dexamethasone	Recommended as preferred therapy with or without remdesivir	Promptly add either agent (rating recommendation/ evidence level: baricitinib AI; tocilizumab BIIa) to dexamethasone ± remdesivir in most patients		For patients without an indication for therapeutic anticoagulation: Prophylactic dose of heparin, unless contraindicated (AI) For patients on therapeutic dose of heparin in non-ICU setting and transferred to ICU, switch to prophylactic dose of heparin, unless another indication for therapeutic anticoagulation (BIII)
Mechanical ventilation or extracorporeal membrane oxygenation requirement	N/A	Recommmeded to be administered to all patients (AI).	Promptly add either agent (Rating recommendation/ Evidence level: baricitinib BIIa; tocilizumab BIIa) to dexamethasone in most patients		

[a]Alternative corticosteroids at equivalent glucocorticoid dose may be considered.

[b]Although approximately 33% of REMAP-CAP and RECOVERY patients received a second dose of tocilizumab at the treating physician's discretion, evidence to support repeat dosing is insufficient.

N/A = No recommendation provided.

Information from: National Institutes of Health (NIH); COVID-19 Treatment Guidelines Panel. Coronavirus Disease 2019 (COVID-19) Treatment Guidelines. Available at www.covid19treatmentguidelines.nih.gov/about-the-guidelines/table-of-contents/. RECOVERY Collaborative Group. Tocilizumab in patients admitted to hospital with COVID-19 (RECOVERY): a randomised, controlled, open-label, platform trial. Lancet 2021;397:1637-45; REMAP-CAP Investigators, Gordon AC, Mouncey PR, et al. Interleukin-6 receptor antagonists in critically ill patients with Covid-19. N Engl J Med 2021;384:1491-502. Ely EW, Ramanan AV, Kartman CE, et al. Efficacy and safety of baricitinib plus standard of care for the treatment of critically ill hospitalised adults with COVID-19 on invasive mechanical ventilation or extracorporeal membrane oxygenation: an exploratory, randomised, placebo-controlled trial. Lancet Respir Med 2022;10:327-36; WHO Solidarity Trial Consortium. Remdesivir and three other drugs for hospitalised patients with COVID-19: final results of the WHO Solidarity randomised trial and updated meta-analyses. Lancet 2022;399:1941-53.

Patient Case

21. S.B. is a 48-year-old woman with severe obesity and heart disease intubated emergently at an outside hospital for refractory hypoxemic respiratory failure despite maximal MV support. S.B. is being evaluated for venovenous ECMO. She is quickly transferred to the MICU for evaluation. S.B. is initiated on dexamethasone 6 mg intravenously once daily, and her analgesia-sedative regimen is modified to maintain deep sedation in anticipation of starting continuous neuromuscular blockade. S.B. is receiving norepinephrine at 10 mcg/minute, which continues to be tapered to maintain a mean arterial pressure greater than 65 mm Hg. S.B. is being considered for remdesivir, which is in ample supply at your institution. Which best describes the current guideline recommendations for remdesivir in patients with COVID-19 requiring MV?

    A. Shown to decrease in-hospital mortality.
    B. Not recommended because of safety concerns.
    C. Recommended to shorten time to clinical recovery.
    D. Uncertain benefit on clinical recovery and mortality.

REFERENCES

Ventilator-Associated Pneumonia

1. American Thoracic Society; Infectious Diseases Society of America. Guidelines for the management of adults with hospital-acquired, ventilator-associated, and healthcare-associated pneumonia. Am J Respir Crit Care Med 2005;171:388-416. Previous evidence-based multi-organizational guidelines for the management of hospital-acquired pneumonia, including VAP. Included to support transition between guidelines.
2. Bouglé A, Tuffet S, Federici L et al. Comparison of 8 versus 15 days of antibiotic therapy for *Pseudomonas aeruginosa* ventilator-associated pneumonia in adults: a randomized, controlled, open-label trial. Intensive Care Med 2022;48:841-49.
3. Chastre J, Fagon JY. Ventilator-associated pneumonia. Am J Respir Crit Care Med 2002;165:867-903. Comprehensive review of the foundational aspects of the diagnosis and management of VAP.
4. Chastre J, Wolff M, Fagon JY, et al.; PneumA Trial Group. Comparison of 8 vs 15 days of antibiotic therapy for ventilator-associated pneumonia in adults: a randomized trial. JAMA 2003;290:2588-98. Pivotal study helping define the optimal duration of definitive antibiotic therapy for VAP. Foundation for current guideline recommendations for duration of antibiotic therapy.
5. Dimopoulos G, Poulakou G, Pneumatikos IA, et al. Short- vs long-duration antibiotic regimens for ventilator-associated pneumonia: a systematic review and meta-analysis. Chest 2013; 144:1759–67.
6. Dudeck MA, Weiner LM, Allen-Bridson K, et al. National Healthcare Safety Network (NHSN) report, data summary for 2012, device-associated module. Am J Infect Control 2013;41:1148-66. Updated CDC-based surveillance data and benchmarks for rates of device-related infections.
7. Johnstone J, Meade M, Lauzier F, et al. Effect of probiotics on incident ventilator-associated pneumonia in critically ill patients: a randomized clinical trial. JAMA 2021;326:1024-33. doi: 10.1001/jama.2021.13355
8. Kalil AC, Metersky ML, Klompas M, et al. Management of Adults With Hospital-acquired and Ventilator-associated Pneumonia: 2016 Clinical Practice Guidelines by the Infectious Diseases Society of America and the American Thoracic Society. Clin Infect Dis. 2016 Sep 1;63(5):e61-e111. Most recent evidence-based multi-organizational guidelines for the management of hospital-acquired pneumonia, including VAP.
9. Klompas M, Branson R, Cawcutt K, et al. Strategies to prevent ventilator-associated pneumonia, ventilator-associated events, and nonventilator hospital-acquired pneumonia in acute-care hospitals: 2022 Update. Infect Control Hosp Epidemiol 2022;43:687-713. doi: 10.1017/ice.2022.88
10. Klompas M, Branson R, Eichenwald EC, et al. Strategies to prevent ventilator-associated pneumonia in acute care hospitals: 2014 update. Infect Control Hosp Epidemiol 2014;35:915-36. Updated evidence-based and expert panel recommendations for preventing VAP. Endorsed by many infection and acute care organizations.
11. Klompas M, Li L, Menchaca JT, et al. Ultra-short-course antibiotics for patients with suspected ventilator-associated pneumonia but minimal and stable ventilator settings. Clin Infect Dis 2017;64:870-6.
12. Kuti EL, Patel AA, Coleman CI. Impact of inappropriate antibiotic therapy on mortality in patients with ventilator-associated pneumonia and blood stream infection: a meta-analysis. J Crit Care 2008;23:91-100. Comprehensive compilation and quantitative assessment of studies evaluating inappropriate empiric antibiotic therapy for VAP.
13. Pugh R, Grant C, Cooke RP, Dempsey G. Short-course versus prolonged-course antibiotic therapy for hospital-acquired pneumonia in critically ill adults. Cochrane Database Syst Rev 2015; 8:Cd007577.
14. Sharif S, Greer A, Skorupski C, et al. Probiotics in critical illness: a systematic review and meta-analysis of randomized controlled trials. Crit Care Med 20221;50:1175-86. doi: 10.1097/CCM.0000000000005580
15. Tamma PD, Aitken SL, Bonomo RA, Mathers AJ, van Duin D, Clancy CJ. Infectious Diseases Society of America 2023 guidance on the treatment of antimicrobial resistant gram-negative infections. Clin Infect Dis 2023:ciad428. doi: 10.1093/cid/ciad428

16. Yoshimura J, Yamakawa K, Ohta Y, et al. Effect of Gram stain-guided initial antibiotic therapy on clinical response in patients with ventilator-associated pneumonia: the GRACE-VAP randomized clinical trial. JAMA Netw Open 2022;5:e226136. doi: 10.1001/jamanetworkopen.2022.6136

Central Line–Associated Bloodstream Infections

1. Buetti N, Marschall J, Drees M, et al. Strategies to prevent central line-associated bloodstream infections in acute-care hospitals: 2022 Update. Infect Control Hosp Epidemiol 2022;43:553-69. doi: 10.1017/ice.2022.87
2. Centers for Disease Control and Prevention. Central Line-Associated Bloodstream (CLABSI) Event. Available at www.cdc.gov/nhsn/pdfs/pscmanual/4psc_clabscurrent.pdf. Provides CDC surveillance definitions for CLABSI.
3. Centers for Disease Control and Prevention. Healthcare-Associated Infections in the United States, 2006-2016: A Story of Progress. Available at www.cdc.gov/hai/data/archive/data-summary-assessing-progress.html.
4. Dudeck MA, Weiner LM, Allen-Bridson K, et al. National Healthcare Safety Network (NHSN) report, data summary for 2012, device-associated module. Am J Infect Control 2013;41:1148-66. Updated CDC-based surveillance data and benchmarks for rates of device-related infections.
5. Marschall J, Mermel LA, Fakih M, et al. Strategies to prevent central line-associated bloodstream infections in acute care hospitals: 2014 update. Infect Control Hosp Epidemiol 2014;35:753-71.
6. Mermel LA, Allon M, Bouza E, et al. Clinical practice guidelines for the diagnosis and management of intravascular catheter-related infection: 2009 update by the Infectious Diseases Society of America. Clin Infect Dis 2009;49:1-45. Evidence-based guidelines for the clinical diagnosis and management of catheter-related infections, including CLABSI.
7. Pronovost P, Needham D, Berenholtz S, et al. An intervention to decrease catheter-related bloodstream infections in the ICU. N Engl J Med 2006;355:2725-32. Pivotal study demonstrating a systematic approach using interdisciplinary education and procedural checklists to decrease CLABSI in critically ill patients.

Influenza

1. Centers for Disease Control and Prevention. Influenza Signs and Symptoms and the Role of Laboratory Diagnostics Available at www.cdc.gov/flu/symptoms/index.html.
2. Centers for Disease Control and Prevention. FluView: A Weekly Influenza Surveillance Report Prepared by the Influenza Division. Available at https://www.cdc.gov/flu/weekly/index.htm.
3. Centers for Disease Control and Prevention. Influenza Antiviral Medications: Summary for Clinicians. Available at www.cdc.gov/flu/professionals/antivirals/summary-clinicians.htm..
4. de Jong MD, Ison MG, Monto AS, Metev H, Clark C, O'Neil B, Elder J, McCullough A, Collis P, Sheridan WP. Evaluation of intravenous peramivir for treatment of influenza in hospitalized patients. Clin Infect Dis 2014;59:e172-85. Clinical trial of intravenous peramivir compared to placebo in hospitalized patients.
5. Flannery AH, Thompson Bastin ML. Oseltamivir dosing in critically ill patients with severe influenza. Ann Pharmacother 2014;48:1011-8. Descriptive review of evidence related to oseltamivir dosing in critically ill patients.
6. Harper SA, Bradley JS, Englund JA, et al. Seasonal influenza in adults and children—diagnosis, treatment, chemoprophylaxis, and institutional outbreak management: clinical practice guidelines of the Infectious Diseases Society of America. Clin Infect Dis 2009;48:1003-32.
7. Rothberg MB, Haessler SD. Complications of seasonal and pandemic influenza. Crit Care Med 2010;38(suppl):e91-e97. Critical care–focused review of severe influenza-related complications.

Catheter-Associated Urinary Tract Infections

1. Centers for Disease Control and Prevention. Catheter-Associated Urinary Tract Infection (CAUTI) Event. Available at www.cdc.gov/nhsn/pdfs/pscmanual/7psccauticurrent.pdf. CDC prevention and surveillance definitions.
2. Daneman N, Rishu AH, Pinto RL, et al.; Canadian Critical Care Trials Group. Bacteremia Antibiotic Length Actually Needed for Clinical Effectiveness (BALANCE) randomised clinical trial: study protocol. BMJ Open 2020;10:e038300.

3. Dudeck MA, Weiner LM, Allen-Bridson K, et al. National Healthcare Safety Network (NHSN) report, data summary for 2012, device-associated module. Am J Infect Control 2013;41:1148-66. Updated CDC-based surveillance data and benchmarks for rates of device-related infections.
4. Hooton RM, Bradley SF, Cardenas DD, et al. Diagnosis, prevention, and treatment of catheter-associated urinary tract infection in adults: 2009 international clinical practice guidelines from the Infectious Diseases Society of America. Clin Infect Dis 2010;50:625-63.
5. Patel P, Advani S, Kofman A, et al. Strategies to prevent catheter-associated urinary tract infections in acute-care hospitals: 2022 update. Infect Control Hosp Epidemiol 2023;44:1209-31. doi: 10.1017/ice.2023

Complicated Intra-abdominal Infection

1. Dellinger RP, Levy MM, Rhodes A, et al. Surviving Sepsis Campaign: international guidelines for management of severe sepsis and septic shock: 2012. Crit Care Med 2013;41:580-637.
2. Levy MM, Dellinger RP, Townsend SR, et al. The Surviving Sepsis Campaign: results of an international guideline-based performance improvement program targeting severe sepsis. Crit Care Med 2010;38:367-74.
3. Marshall JC, Innes M. Intensive care unit management of intra-abdominal infection. Crit Care Med 2003;31:2228-37. In-depth descriptive review of pathophysiology and management strategies of complicated intra-abdominal infection in critically ill patients.
4. Mazuski JE, Tessier JM, May AK, et al. The Surgical Infection Society Revised Guidelines on the Management of Intra-Abdominal Infection. Surgical Infections 2017;18:1-76. Updated guidelines from the Surgical Infection Society. Previous guidelines jointly included IDSA.
5. Sartelli M, Coccolini F, Kluger Y, et al. WSES/GAIS/SIS-E/WSIS/AAST global clinical pathways for patients with intra-abdominal infections. World J Emerg Surg 2021;16:49.
6. Sawyer RG, Claridge JA, Nathens AB, et al. Trial of short-course antimicrobial therapy for intraabdominal infection. N Engl J Med 2015;372:1996-2005.
7. Solomkin JS, Mazuski JE, Bradley JS, et al. Diagnosis and management of complicated intra-abdominal infection in adults and children: guidelines by the Surgical Infection Society and the Infectious Diseases Society of America. Clin Infect Dis 2010;50:133-64.

Acute Pancreatitis

1. Banks PA, Bollen TL, Dervenis C, et al. Classification of acute pancreatitis – 2012: revision of the Atlanta classification and definitions by international consensus. Gut 2013;62:102-11.
2. Crockett SD, Wani S, Gardner TB, et al. American Gastroenterological Association Institute guideline on initial management of acute pancreatitis. Gastroenterology 2018;154:1096-101.
3. Dellinger EP, Tellado JM, Soto NE, et al. Early antibiotic treatment for severe acute necrotizing pancreatitis: a randomized, double-blind, placebo-controlled Study. Ann Surg 2007;245:674-83.
4. de-Madaria E, Buxbaum JL, Maisonneuve P, et al. Aggressive or moderate fluid rescucitation in acute pancreatitis. N Engl J Med 2022;387:989-1000.
5. Frossard JL, Steer ML, Pastor CM. Acute pancreatitis. Lancet 2008;371:143-52.
6. Leppäniemi A, Tolonen M, Tarasconi A, et al. 2019 WSES guidelines for the management of severe acute pancreatitis. World J Emerg Surg 2019;14:27.
7. Mazuski JE, Tessier JM, May AK, et al. The Surgical Infection Society revised guidelines on the management of intra-abdominal infection. Surgical Infections 2017;18:1-76. Additional set of guidelines that address prophylactic antibiotic therapy in patients with acute necrotizing pancreatitis.
8. Nathens AB, Curtis JR, Beale RJ, et al. Management of the critically ill patient with severe acute pancreatitis. Crit Care Med 2004;32:2524-36.
9. Tenner S, Baillie J, DeWitt J, et al. American College of Gastroenterology guideline: management of acute pancreatitis. Am J Gastroenterol 2013;108:1400-15.
10. Villatoro E, Mulla M, Larvin M. Antibiotic therapy for prophylaxis against infection of pancreatic necrosis in acute pancreatitis. Cochrane Database Syst Rev 2012;5:CD002941.
11. Whitcomb DC. Acute pancreatitis. N Engl J Med 2006;354:2142-50.

***Clostridioides difficile* Infection**

1. Apisarnthanarak A, Razavi B, Mundy LM. Adjunctive intracolonic vancomycin for severe *Clostridium difficile* colitis: case series and review of the literature. Clin Infect Dis 2002;35:690-6.
2. Johnson S, Lavergne V, Skinner AM, et al. Clinical practice guideline by the Infectious Diseases Society of America (IDSA) and Society for Healthcare Epidemiology of America (SHEA): 2021 focused update guidelines on management of Clostridioides difficile infection in adults. Clin Inf Dis 2021. Available at www.idsociety.org/practice-guideline/clostridioides-difficile-2021-focused-update/.
3. Kelly CP, LaMont JT. *Clostridium difficile*—more difficult than ever. N Engl J Med 2008;359:1932-40.
4. Kelly CR, Fischer M, Allegretti JP, et al. ACG Clinical Guidelines: prevention, diagnosis, and treatment of *Clostridioides difficile* infections, Am J Gastroenterol 2021;116:1124-47.
5. Kociolek L, Gerding D, Carrico R, et al. Strategies to prevent *Clostridioides difficile* infections in acute-care hospitals: 2022 update. Infect Control Hosp Epidemiol 2023;44:527-49. doi: 10.1017/ice.2023.18.
6. McDonald LC, Gerding DN, Johnson S, et al. Clinical practice guidelines for *Clostridium difficile* infection in adults and children: 2017 update by the Infectious Diseases Society of America (IDSA) and Society for Healthcare Epidemiology of America (SHEA). Clin Infect Dis 2018;66:987-94.
7. Owens RC. *Clostridium difficile*–associated disease: an emerging threat to patient safety. Pharmacotherapy 2006;26:299-311.
8. Surawicz CM, Brandt LJ, Binion DG, et al. Guidelines for Diagnosis, Treatment, and Prevention of *Clostridium difficile* Infections. Am J Gastroenterol 2013;108:478–98.
9. Zar FA, Bakkanagari SR, Moorthi KMLST, et al. A comparison of vancomycin and metronidazole for the treatment of *Clostridium difficile*–associated diarrhea, stratified by disease severity. Clin Infect Dis 2007;45:302-7.

Wound Infection

1. Anaya DA, Dellinger EP. Necrotizing soft-tissue infection: diagnosis and management. Clin Infect Dis 2007;44:705-10.
2. Calderwood M, Anderson D, Bratzler DW, et al. Strategies to prevent surgical site infections in acute-care hospitals: 2022 update. Infect Control Hosp Epidemiol 2023;44:695-720. doi: 10.1017/ice.2023.67
3. Lappin E, Ferguson AJ. Gram-positive toxic shock syndromes. Lancet Infect Dis 2009;9:281-90.
4. May AK, Stafford RE, Bulger EM, et al. Treatment of complicated skin and soft tissue infections: Surgical Infection Society guidelines. Surg Infect 2009;10:467-99.
5. Sartelli M, Guirao X, Hardcastle TC, et al. 2018 WSES/SIS-E consensus conference: recommendations for the management of skin and soft-tissue infections. World J Emerg Surg 2018;13:58.
6. Stevens DL, Bisno AL, Chambers HF, et al. Practice guidelines for the diagnosis and management of skin and soft tissue infections: 2014 update by the Infectious Diseases Society of America. Clin Infect Dis 2014;59:e10-e52.

Severe Nonthermal Cutaneous Injury

1. Bastuji-Garin S, Rzany B, Stern RS, et al. Clinical classification of cases of toxic epidermal necrolysis, Stevens-Johnson syndrome, and erythema multiforme. Arch Dermatol 1993;129:92-6.
2. Creamer D, Walsh S, Smith C, et al. U.K. guidelines for the management of Stevens-Johnson syndrome/toxic epidermal necrolysis in adults 2016. British Journal Dermatology 2016;174:1194-1227.
3. Ely EW, Ramanan AV, Kartman CE, et al. Efficacy and safety of baricitinib plus standard of care for the treatment of critically ill hospitalised adults with COVID-19 on invasive mechanical ventilation or extracorporeal membrane oxygenation: an exploratory, randomised, placebo-controlled trial. Lancet Respir Med 2022;10:327-36.
4. Huang YC, Li YC, Chen TJ. The efficacy of intravenous immunoglobulin for the treatment of toxic epidermal necrolysis: a systematic review and meta-analysis. British Journal Dermatology 2012;167:424-32.
5. Kumar Gupta L, Mani Martin A, Argawai N, et al. Guidelines for the management of Stevens–Johnson syndrome/toxic epidermal necrolysis: an Indian perspective. Indian J Dermatol Venereol Leprol 2016;82:603-25.

6. Palmieri TL, Greenhalgh DG, Saffle JR, et al. A multicenter review of toxic epidermal necrolysis treated in U.S. burn centers at the end of the twentieth century. J Burn Care Rehabil 2002;23:87-96.
7. Pereira FA, Mudgil AV, Rosmarin DM. Toxic epidermal necrolysis. J Am Acad Dermatol 2007;56:181-200.
8. Seminario-Vidal L, Kroshinsky D, Malachowski SJ, et al. Society of Dermatology Hospitalists supportive care guidelines for the management of Stevens-Johnson syndrome/toxic epidermal necrolysis in adults. J Am Acad Dermatol 2020;82:1553-67.
9. Struck MF, Hilbert P, Mockenhaupt M, et al. Severe cutaneous adverse reactions: emergency approach to non-burn epidermolytic syndromes. Intensive Care Med 2010;36:22-32.

Severe Acute Respiratory Syndrome Coronavirus 2

1. Alhazzani W, Moller MH, Arabi YN, et al. Surviving Sepsis Campaign: guidelines on the management of critically ill adults with coronavirus disease 2019 (COVID-19). Crit Care Med 2020;48:e440-e469.
2. Beigel JH, Tomashek KM, Dodd LE. Remdesivir for the treatment of COVID-19: preliminary report. N Engl J Med 2020;383:994.
3. Kalil AC, Patterson TF, Mehta AK, et al. Baricitinib plus remdesivir for hospitalized adults with Covid-19. N Engl J Med 2021;384:795-807.
4. Marconi VC, Ramanan AV, de Bono S, et al. Efficacy and safety of baricitinib for the treatment of hospitalised adults with COVID-19 (COV-BARRIER): a randomised, double-blind, parallel-group, placebo-controlled phase 3 trial. Lancet Resp Med 2021. Available at www.thelancet.com/journals/lanres/article/PIIS2213-2600(21)00331-3/fulltext.
5. National Institutes of Health (NIH); COVID-19 Treatment Guidelines Panel. Coronavirus Disease 2019 (COVID-19) Treatment Guidelines. Available at www.covid19treatmentguidelines.nih.gov/about-the-guidelines/table-of-contents/.
6. RECOVERY Collaborative Group; Horby P, Lim WS, et al. Dexamethasone in hospitalized patients with COVID-19N Engl J Med 2021;384:693-704.
7. RECOVERY Collaborative Group. Tocilizumab in patients admitted to hospital with COVID-19 (RECOVERY): a randomised, controlled, open-label, platform trial. Lancet 2021;397:1637-45.
8. REMAP-CAP Investigators, Gordon AC, Mouncey PR, et al. Interleukin-6 receptor antagonists in critically ill patients with Covid-19. N Engl J Med 2021;384:1491-502.
9. Rosas IO, Baru N, Waters M, et al. Tocilizumab in hospitalized patients with severe Covid-19 pneumonia. N Engl J Med 2021;384:1503-16.
10. WHO Rapid Evidence Appraisal for COVID-19 Therapies (REACT) Working Group. Association between administration of systemic corticosteroids and mortality among critically ill patients with COVID-19: a meta-analysis. JAMA 2020;324:1330-41.
11. WHO Solidarity Trial Consortium. Remdesivir and three other drugs for hospitalised patients with COVID-19: final results of the WHO Solidarity randomised trial and updated meta-analyses. Lancet 2022;399:1941-53.
12. Wiersinga WJ, Rhodes A, Cheng AC, et al. Pathophysiology, transmission, diagnosis, and treatment of coronavirus disease 2019 (COVID-19): a review. JAMA 2020;324:782-93.

ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: A**

The patient has suspicion for MDR VAP, as evidenced by the presence of clinical signs of infection, the patient's increased sputum production, and the patient's having been in the ICU for 5 days or longer. Empiric antibiotic therapy should be initiated after obtaining a respiratory culture and be based on patient-specific risk factors for MDROs, together with local pathogen prevalence and antibiotic susceptibility, to increase the likelihood of providing timely appropriate antibiotic therapy (Answer A is correct). Gram stain results can be used to help guide empiric therapy as per GRACE-VAP, but antibiotics should not be delayed 1–2 days awaiting Gram stain, preliminary or final respiratory culture results, as well as blood or urine cultures, may cause an unacceptable delay in appropriate antibiotic therapy (Answers B–D are incorrect).

2. **Answer: B**

Although this patient is suspected of having early-onset VAP for the current admission, a history of recent intravenous antibiotic therapy is a risk factor for MDROs. Empiric antibiotic therapy for VAP in patients with MDRO risk factors should include agents active against *P. aeruginosa* and MRSA. Empiric combination therapy against *P. aeruginosa* is recommended to increase the likelihood of appropriate antibiotic therapy (Answer B is correct; Answers A and C are incorrect) with a β-lactam antibiotic as one of the preferred agents (Answer D is incorrect). Atypical bacteria coverage is not necessary because their prevalence is low, although consideration should be given if there is a poor response to initial therapy.

3. **Answer: B**

Based on the PneumA trial and related meta-analyses, the most recent IDSA VAP guidelines recommend definitive antibiotic therapy duration of 7 full treatment days for all patients. This is further emphasized in this patient, who has VAP caused by *Klebsiella* spp., which are lactose-fermenting gram-negative bacilli, and who received appropriate empiric antibiotic therapy and had an appropriate clinical response during therapy (Answer B is correct; Answers A, C, and D are incorrect).

4. **Answer: D**

In the absence of other suspected sources (i.e., no change in chest radiograph), CLABSI should be suspected as the cause of new-onset fever and leukocytosis, given the emergency placement and related duration of the CVC (Answer B is incorrect). Although catheter removal should strongly be considered, cultures should be obtained before catheter removal for documentation if the patient has a bloodstream infection (Answer C is incorrect). Initiation of antibiotic therapy should be considered, if appropriate, but only after cultures of the suspected source are obtained (Answer D is correct; Answer A is incorrect).

5. **Answer: D**

This patient, who is thought to have a CLABSI, has risk factors for MDROs, given that the patient was hospitalized for 5 or more days. Empiric antibiotic therapy choices should include agents active against MRSE and MRSA as well as *P. aeruginosa* (Answer D is correct; Answer A is incorrect). Linezolid is active against MRSA; however, it should be considered only for definitive therapy because its empiric use in patients with a CLABSI is associated with worse outcomes (Answer C is incorrect). Fluconazole may be considered in addition to antibiotic therapy, but monotherapy is not recommended empirically (Answer B is incorrect).

6. **Answer: B**

The guideline recommendation for definitive antibiotic therapy duration is 7–14 days from the first negative blood culture in patients with uncomplicated gram-negative CLABSI. Longer durations of therapy should be considered in patients with persistent bacteremia or if the patient has a poor clinical response. (Answer B is correct; Answers A and C are incorrect). Patients with complicated bacteremia (e.g., endocarditis, septic thrombus, chronic intravascular hardware) should receive 4-6 weeks of therapy (Answer D is incorrect).

7. **Answer: C**

The presence of new fever and an elevated WBC in conjunction with an indwelling urinary catheter and pyuria on urinalysis is highly suggestive of a CAUTI rather than worsening pneumonia (Answer B is incorrect). Similar to other ICU-related infections, empiric

antibiotic therapy should be considered if there is strong clinical suspicion for infection (Answer C is correct). Waiting for final culture results before initiating antimicrobial therapy in patients thought to have a CAUTI may delay appropriate treatment (Answer A is incorrect). Although it is recommended to discontinue the urinary catheter and replace as necessary, a confirmatory urinalysis is not needed if the catheter has been indwelling for less than 2 weeks (Answer D is incorrect).

8. **Answer: A**

The causative pathogens associated with a CAUTI, with or without multidrug-resistant risk factors, are more heterogeneous than an uncomplicated or community-acquired urinary tract infection. Nonetheless, *E. coli* remains the most common bacterial pathogen together with other gram-negative bacilli and *Candida* spp. (Answer A is correct). Except for *S. maltophilia* (Answer B is incorrect), other associated pathogens include those listed in Answers B–D; however, the absence of *E. coli* makes these answers incorrect.

9. **Answer: D**

This patient has complicated intra-abdominal infection from secondary peritonitis caused by colonic perforation. Although it is community acquired, the presence of septic shock suggests severe classification increasing the risk of gram-negative MDROs (Answer C is incorrect). The involvement of the colon also obligates antibiotic therapy active against anaerobes and enterococci (Answer B is incorrect); MRSA is an unlikely pathogen (Answer A is incorrect). According to this, piperacillin/tazobactam is the most appropriate agent listed (Answer D is correct).

10. **Answer: A**

The approach to managing a complicated intra-abdominal infection includes timely source control and initiation of empiric antimicrobial therapy active against likely pathogens. This involves assessment of the anatomic location of the source, risk factors for multidrug-resistant organisms (i.e., community- or health care–associated disposition), and severity of illness. This patient has community-acquired secondary peritonitis from a distal small bowel perforation with concomitant septic shock. Although his disease is community acquired, the presence of septic shock suggests a severe classification, increasing the risk of gram-negative multidrug-resistant organisms and involvement of the distal small bowel obligates empiric coverage for anaerobic pathogens and enterococci, for all of which piperacillin/tazobactam should provide appropriate empiric coverage (Answer A is correct). Ertapenem would likely have adequate empiric coverage for enteric, gram-negative pathogens, both aerobic and anaerobic; however, lack or enterococcal coverage makes it less optimal (Answer B is incorrect). Ciprofloxacin lacks sufficient activity against enterococci or anaerobic pathogens, and MRSA coverage with vancomycin is unnecessary (Answer C is incorrect). Cefazolin is too narrow as empiric therapy for a severe community-acquired complicated intra-abdominal infection, and fluconazole should be reserved for definitive therapy for fluconazole-susceptible yeast identified on final culture (Answer D is incorrect).

11. **Answer: D**

Acute necrotizing pancreatitis often presents with signs and symptoms consistent with systemic inflammatory response syndrome. However, most cases of severe necrotizing pancreatitis are sterile and do not require antibiotic therapy. The presence of a pancreatic pseudocyst or abscess increases the likelihood of infected necrotizing pancreatitis, which thus warrants empiric antimicrobial therapy. However, this patient has no evidence of these on CT scan; thus, empiric antibiotic therapy is not indicated (Answers A–C are incorrect). Moreover, prophylactic antimicrobial therapy is not recommended in the recent guidelines, despite the presentation of systemic inflammation and pancreatic necrosis (Answer D is correct). A prospective, randomized, noninferiority study most strongly supports these recommendations.

12. **Answer: D**

This patient has acute, severe pancreatitis with CT evidence of pancreatic abscess. Gram stain of fluid obtained from CT-guided drainage suggests the presence of gram-negative bacilli and yeast (Answer A is incorrect). Empiric antimicrobial therapy is indicated for suspected infected pancreatitis. Extended-spectrum carbapenems, which achieve relevant pancreatic fluid concentrations, are effective in the management of infected pancreatitis. Addition of anti-candidal therapy is indicated, given the presence of yeast on the Gram stain (Answer D is correct; Answers B and C are incorrect).

13. **Answer: B**
This patient is thought to have fulminant (IDSA 2017) CDI, given his recent exposure to broad-spectrum antibiotic therapy, signs of infection, CT findings, and the presence of hypotension. Combination therapy with metronidazole and enteral vancomycin is indicated (Answers A and D are incorrect). The presence of ileus requires consideration of adding intracolonic vancomycin because of possible impaired delivery to the colon through enteral routes (Answer B is correct; Answer C is incorrect).

14. **Answer: D**
The patient in this case presents with what is likely an initial-episode, severe CDI, as supported by new-onset diarrhea and clinical signs and symptoms of infection. Of note, the patient is not hypotensive and has a lactate below 2–4 mmol/L, suggesting the absence of shock or other defining features of fulminant CDI; thus, monotherapy with enteral vancomycin is the best available answer (Answer D is correct). Although the 2021 IDSA/SHEA focused guideline update recommends fidaxomicin for non-fulminant, first-episode CDI, oral/enteral vancomycin is an acceptable alternative, especially if fidaxomicin is unavailable. Combination therapy with oral/enteral vancomycin and intravenous metronidazole should be reserved for a complicated CDI (Answer C is incorrect), and intravenous metronidazole is not recommended (Answer B is incorrect). The combination of metronidazole and intracolonic is recommended for patients with a complicated CDI and concern for ileus or inability to deliver oral/enteral vancomycin to the colon (Answer A is incorrect).

15. **Answer: C**
Necrotizing fasciitis is a severe, life-threatening infection that is often polymicrobial. Prompt surgical debridement of necrotic tissue and broad-spectrum antibiotic therapy are the mainstays of initial therapy. Agents active against *S. pyogenes*, MRSA, and aerobic and anaerobic gram-negative bacilli should be initiated empirically, together with adjunctive clindamycin added, which may decrease bacterial toxin production. The empiric regimen of piperacillin/tazobactam, vancomycin, and clindamycin is most appropriate, given the severity of infection despite the absence of traditional multidrug-resistant organism risk factors (Answer C is correct). Penicillin G is not broad enough (Answer B is incorrect), and the combination of ceftaroline and vancomycin does not contain clindamycin (Answer A is incorrect). Combination of clindamycin and linezolid do not include empiric gram-negative coverage. (Answer D is incorrect).

16. **Answer: B**
The intraoperative cultures of this patient's necrotizing infection are of concern for *S. pyogenes*. *S. pyogenes* produces exotoxin, which is associated with tissue necrosis. Although bactericidal antibiotic therapy is necessary for eradicating *S. pyogenes*, adjunctive clindamycin may decrease toxin production, which could limit the extent of tissue necrosis (Answer B is correct). Vancomycin should be reserved for patients with an *S. pyogenes* infection who have a β-lactam allergy because β-lactam resistance is rare (Answer C is incorrect). Moreover, synergistic antibiotic therapy with gentamicin is not needed, given the effectiveness of gram-positive β-lactams against *S. pyogenes* (Answer A is incorrect). The role of IVIG in streptococcal necrotizing fasciitis is controversial. A small randomized, placebo-controlled trial – stratified on the basis of need for surgery and clindamycin treatment – showed no improvement in survival or reduction in the time to no further progression of necrotizing fasciitis or cellulitis in 21 patients with streptococcal toxic shock syndrome. As such, the guidelines do not recommend IVIG until additional studies are available (Answer D is incorrect).

17. **Answer: A**
One of the cornerstones of managing severe cutaneous injury includes volume resuscitation, preferentially with crystalloids (Answer A is correct). The usefulness of immunomodulating therapies such as corticosteroids and cyclophosphamide is limited by observational and poorly controlled evidence or case reports; thus, use of immunomodulating therapies should be reserved for specialty centers using formal protocols with consideration for treatment under clinical study or registry (Answers B and D are incorrect). Wound care is imperative in these patients; however, adding unnecessary drugs that could confound response or worsen the injury (e.g., antibiotics) should be avoided (Answer C is incorrect) unless there is an objectively suspected or confirmed infection.

18. **Answer: B**
Use of IVIG for Stevens-Johnson syndrome (SJS)/TEN is controversial. Available evidence is from case reports/series, observational cohort studies, or small, single-center randomized trials (Answer C is incorrect). Because of bias, limited external validity, and mixed results/observations between publications, meta-analyses and consensus guidelines do not broadly endorse the use of IVIG (Answer D is incorrect). The available data analyses that suggest that IVIG decreases the SCORTEN-related mortality in patients with TEN and a middle to higher SCORTEN score; there are even fewer data for the role of IVIG in SJS (Answer A is incorrect). Given the limited evidence supporting pharmacotherapeutic interventions and the life-threatening severity of TEN, burn referral centers should base the decision to use IVIG on local protocols/guidelines developed by interdisciplinary practitioners who have objectively evaluated the available evidence regarding safety, efficacy, and pharmacogenomic implications (Answer B is correct).

19. **Answer: D**
This patient likely has severe influenza amid a local seasonal outbreak. Local infection patterns suggest a prevalence of influenza A and B strains. Empiric influenza-specific therapy against these strains should be initiated in patients with severe influenza before confirmatory test results are known to avoid a delay in appropriate therapy (Answer B is incorrect). Neuraminidase-based therapy is recommended for modern influenza A and B stains (Answer D is correct; Answer C is incorrect). Even if they are outside 48 hours from symptom onset, patients with severe influenza have benefited from therapy initiated beyond this period (Answer A is incorrect).

20. **Answer: B**
Neuraminidase inhibitors are the mainstay of treatment because resistance to adamantanes has developed in contemporary influenza A strains, and these agents have no antiviral activity against influenza B (Answer A is incorrect). Inhaled zanamivir is not appropriate for administration to mechanically ventilated patients (Answer C is incorrect). Intravenous peramivir also is available; however, data are limited for its use in critically ill patients with severe, complicated influenza infection (Answer D is incorrect). Enteral oseltamivir is the preferred therapy for current influenza B strains in critically ill patients with severe, complicated influenza infection (Answer B is correct). Clinicians should be mindful of the need for renal dosage adjustment, if necessary.

21. **Answer: D**
Remdesivir is recommended for patients requiring supplemental oxygen. On the basis of the results of the ACTT study showing no difference in recovery time or mortality in more severe subgroups, the NIH guidelines do not recommend for or against remdesivir in patients requiring high-flow oxygen, MV, or ECMO (Answer D is correct; Answers A and C are incorrect). If remdesivir is initiated before advancing to MV, it is recommended to continue and complete therapy. Available evidence guidelines do not suggest increased adverse effects with remdesivir between less and more severely ill patients (Answer B is incorrect).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: D**
The patient described has early onset VAP without apparent updated risk factors for MDROs (Answers A, B, and C, are incorrect). Pathogens associated with early onset hospital-acquired pneumonia and VAP in the absence of other MDR risk factors are usually community-acquired organisms, including *S. pneumoniae*, MSSA, *H. influenzae*, and enteric gram-negative bacilli (Answer D is correct). However, because some epidemiological studies have documented the occurrence of traditionally nosocomial or late-onset pathogens within 5 days from admission, guideline updates recommend that *P. aeruginosa* coverage be considered as empiric therapy for all patients suspected of having VAP. This further emphasizes the importance of understanding local VAP pathogen trends amid MDR risk factors and time from admission to onset. Atypical bacteria are rarely associated with early onset VAP. If MDRO risk factors were present, MRSA and *P. aeruginosa* would also be considered.

2. **Answer: B**
Empiric antibiotic choices for VAP should be based on the likely causative pathogens, the presence of MDRO risk factors, and local antibiotic susceptibility patterns. The 2016 IDSA guidelines recommend *P. aeruginosa* coverage as empiric therapy for all patients suspected of having VAP, with monotherapy antipseudomonal β-lactam sufficient in the absence of MDRO risk factors or if the local antibiogram suggests less than 10% resistance. Therefore, cefepime is preferred to ceftriaxone (Answer B is correct; Answers C and D are incorrect). Atypical bacteria are rarely associated with early-onset VAP (Answer A is incorrect).

3. **Answer: D**
Gram-positive organisms are the most common cause of CLABSI, including MRSE and MRSA. Vancomycin is the best option listed for empiric management (Answer D is correct). Although linezolid has a sufficient spectrum of activity against these organisms, it is not recommended for empiric management of CLABSI because of concerns for worse patient outcomes (Answer B is incorrect). The other options are inactive against MRSE and MRSA and could be considered in addition to vancomycin if there were a high suspicion for additional pathogens (Answers A and C are incorrect).

4. **Answer: D**
Given the high prevalence of the 2009 H1N1 subtype of influenza A, oseltamivir is the empiric drug of choice. In addition, oseltamivir has high-level activity against other contemporary influenza A and B subtypes (Answer D is correct). Zanamivir also has high-level activity against these strains; however, inhaled therapy through the mechanical ventilator is not indicated because of insufficient systemic delivery, and intravenous zanamivir is available through compassionate use and indicated only if oseltamivir cannot be administered (e.g., patient is unable to receive enteral medications, has poor absorption) (Answers B and C are incorrect). Amantadine has insufficient activity against most contemporary influenza A and B strains (Answer A is incorrect).

5. **Answer: B**
Although a health care–associated CAUTI is caused by a more diverse spectrum of pathogens, *E. coli* is still the most common pathogen and is responsible for around 30% of cases (Answer B is correct). The other pathogens listed are also possible and should be considered when choosing empiric antibiotic therapy in patients with a suspected CAUTI (Answers A, C, and D are incorrect).

6. **Answer: D**
This patient has community-acquired complicated intra-abdominal infection involving the middle small intestine. Although enteric gram-negative bacilli (e.g., *E. coli*, *Klebsiella* spp.) are the most common pathogens related to this type of infection, patients with severe disease, as evidenced by concomitant septic shock, are at a higher risk of MDROs, including *P. aeruginosa* and enterococci. Piperacillin/tazobactam has empiric activity against these organisms, whereas the other regimens/agents listed have relevant gaps in the bacterial spectrum relative to these pathogens (Answer D is correct; Answers A–C are incorrect).

7. **Answer: A**
This patient presents with severe acute pancreatitis with radiographic evidence of pancreatic necrosis. Although the patient presents with SIRS, the absence of significant fluid collection or abscess suggests there is no concomitant infection. As such, there is no indication for empiric antibiotic therapy at this time (Answer A is correct; Answer B is incorrect). Most recent evidence

and guideline recommendations do not support prophylactic antibiotic therapy for preventing infection of the necrotic tissue (Answer D is incorrect). The mainstay of therapy for this patient is volume resuscitation and consideration of surgical debridement of pancreatic necrosis if persistent SIRS is evident (Answer C is incorrect).

8. **Answer: A**
Metronidazole orally or through a feeding tube is not recommended for management of an initial-episode CDI (Answer B is incorrect). Intravenous metronidazole is indicated for fulminant CDI in combination with vancomycin (Answer C is incorrect) In the absence of fulminant CDI (i.e., shock, megacolon, ileus), the 2021 IDSA focused guideline update recommends fidaxomicin over oral/enteral vancomycin for first-episode CDI to increase the likelihood of complete clinical response (Answer A is correct; Answer D is incorrect). If fidaxomicin is unavailable or cost-prohibitive, oral/enteral vancomycin is an acceptable alternative.

9. **Answer: A**
This patient has a superficial incisional wound infection requiring opening and the local debridement of infected material. Lack of systemic signs of infection and no involvement of the fascia suggest that no antibiotic therapy is necessary at this time (Answer A is correct; Answers B–D are incorrect). If the infection extends to include these features or worsened erythema consistent with cellulitis, empiric antibiotic therapy may be warranted.

10. **Answer: B**
Similar to severe thermal cutaneous injury, the foundation for managing TEN is volume resuscitation and supportive care as well as removal of all suspected causes (Answer B is correct). The utility of corticosteroids is limited and may be harmful to wound healing (Answer A is incorrect). Addition of unnecessary drugs that could confound response or worsen the injury (e.g., antibiotics) should be avoided (Answers C and D are incorrect).

11. **Answer: A**
The RECOVERY trial showed that dexamethasone decreases 28-day mortality compared with usual care in hospitalized patients with COVID-19 requiring supplemental oxygen or MV. A meta-analysis of available trials also supports corticosteroid use (Answer A is correct). Hydroxychloroquine has limited to no benefit in patients hospitalized with COVID-19 (Answer B is incorrect). Evidence from the ACTT study showed that remdesivir may shorten the duration to clinical recovery, but it has no difference on mortality compared with placebo. Non-controlled, aggregated data have suggested an association with lower mortality, but this has not been confirmed in controlled trials; therefore, remdesivir should be considered in addition to dexamethasone rather than monotherapy in patients with increasing oxygen requirement (Answer C is incorrect). Tocilizumab, an anti–IL-6 monoclonal antibody, can be considered in addition to dexamethasone with or without remdesivir in patients recently hospitalized and demonstrating rapid decline in oxygenation and systemic inflammation. (Answer D is incorrect).